

The Strategic Secretary Problem

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In the classical secretary problem, secretaries have no decision-making agency and are assumed to arrive in a random order. Motivated by modern online labor platforms, we consider a novel variant in which secretaries are strategic agents who choose their arrival times after privately observing their own quality scores. We model the interaction between a hiring firm and $N + 1$ strategic secretaries as a game and characterize the structure of its equilibria. This paper addresses two main questions. First, how do strategic secretaries choose their arrival times in equilibrium? Second, does granting secretaries strategic agency improve or reduce the firm’s ability to select the best candidate? We show that when the firm can commit to a hiring policy and observe secretaries’ quality scores, strategic behavior can improve the firm’s probability of hiring the best secretary, this probability approaches approximately 0.71 as the applicant pool grows, compared with 0.58 in the classical problem. However, when the firm observes only rank-based information about applicants, the probability of hiring the best secretary under strategic arrival is the same as in the classical problem, regardless of the firm’s commitment power; and when the firm observes quality scores but cannot commit ex ante, no symmetric pure-strategy equilibrium exists. Our analysis highlights that granting applicants greater strategic agency can improve hiring outcomes, but only when accompanied by appropriate institutional conditions. In particular, firms should strengthen their ability to commit to hiring policies and prioritize access to more accurate quality signals.

Keywords: Secretary problem, Sequential search, Stackelberg game, Online labor platforms

1. Introduction

The secretary problem is a canonical model of sequential decision-making under uncertainty. In the typical description of this problem, a fixed number of applicants are competing for a single secretarial position offered by a firm. The applicants arrive in a random order, and the hiring firm makes an irrevocable accept-or-reject decision at the time of each arrival, with the objective of selecting the best candidate overall. Once rejected, applicants cannot be recalled. In one of the earliest published versions of this problem (Lindley 1961), the firm perceives each applicant’s relative rank when they arrive, and there is a very simple structure governing the firm’s optimal hiring policy.

Since the earliest study of the secretary problem, there have been numerous extensions and variations of the classical model. As we will review in §2, these extensions fall broadly into three groups, varying the core assumptions about the *firm*, which are: variations to its (a) objectives, (b) information structure, and (c) set of actions. In contrast, there is very limited work that studies the *secretaries'* objectives. For the most part, studies in the literature have modeled applicants as little more than mindless automatons who simply show up in random sequence to be assessed by the firm, who is also the sole decision-maker in the model. These twin assumptions (random-ordering and applicants' lack of agency) have been fundamental in deriving many of the results from the canonical model and its extensions.

Our present paper seeks to study a novel variation of the secretary problem that addresses this gap in the literature. Apart from our motivation to close a theoretical gap, we are also motivated by the dynamics of modern online labor platforms. For context, we argue that the classical secretary problem already captures some high-level strategic tradeoffs inherent in such markets. For instance, a documented key challenge with engaging workers on such online platforms is that their underlying quality can be highly uncertain at the time of hiring (Kokkodis and Ransbotham 2023). While the worker may have an accurate understanding of their own abilities, the hiring firm must rely on limited information, such as past reviews, portfolio samples, and self-reported credentials, to assess the worker's suitability for the task. Further, on many such platforms, firms often review applicants *sequentially*, evaluating candidates one at a time as proposals arrive rather than waiting for the full pool to be known. For instance, on Upwork, once a client posts a job, proposals begin arriving within hours, and the client reviews each applicant's profile and experience with the option to hire, decline, or archive each candidate as proposals come in (Upwork 2024). Therefore, the classical problem, which captures the sequential revelation of uncertain applicant quality, provides a decent foundational structure for modeling such markets.

However, the lack of applicant agency in the classical secretary problem is a tension that is increasingly divergent from how such platforms operate. On such platforms, applicants are anything but passive: They actively choose *when* to submit their proposals within a publicly visible application window, making strategic arrival timing a key lever for influencing hiring outcomes. This is particularly salient on platforms with large applicant

pools, where the timing of one’s application can meaningfully affect one’s chances of standing out. Indeed, platforms themselves actively encourage such strategic thinking: Upwork, for example, provides online resources advising applicants on how to time their applications (Moorhead 2024). As a concrete illustration, once a firm posts a job on Upwork, potential applicants can view the application window and may deliberately decide when to submit, perhaps based on their level of expertise or confidence in their suitability for the role. Consequently, the order in which applications are observed by firms need not be random but can reflect the strategic behavior of applicants, fundamentally reshaping the nature of the selection problem.

Motivated by this tension, in this paper, we study what we will refer to as the *strategic secretary problem*, which is a game-theoretic extension of the canonical problem, where the key twist in our model is that applicants will be endowed with strategic agency. Specifically, we model the interaction between the hiring firm and the pool of applicants as a Stackelberg game: The firm first commits to a hiring policy while anticipating applicants’ best responses, and applicants strategically choose their time of application, which, in turn, will determine the sequence in which they are reviewed by the firm. Within this framework, we analyze the applicants’ equilibrium arrival strategies under different information structures available to the firm and derive the firm’s optimal hiring policy in each case. In the latter part of the paper, we also discuss a simultaneous move variant of our model.

Our objective in this paper is to isolate a high-level strategic tension that is salient for an online labor marketplace but that has been given little attention in the literature, and to understand its ramifications through careful analysis. We freely acknowledge that this paper does *not* develop a high-fidelity model of an online labor platform, nor does it prescribe specific tactical actions under detailed operational conditions for such a platform.

To this end, we view the principal contributions of this paper as two-fold: First, this paper develops and analyzes a variant of the secretary problem in which “secretaries” have strategic agency over their arrival times. To our knowledge, this is a novel variant and departs significantly from the literature which has largely focused on variations to the firm. Second, our analyses delineate conditions under which providing strategic agency is valuable to the hiring firm. As we will show in this paper, the value of agency depends

critically on (a) the ability of the firm to commit to a hiring policy as well as (b) its ability to accurately assess applicant quality upon review. These findings have practical implications for operators of online labor platforms who aim to help their customers (hiring firms) find better matches: providing online platform workers tools to enhance their strategic timing can help with this endeavor, but this is not sufficient and must also be accompanied by tools for customers to better assess candidates. More broadly, our results illustrate a general mechanism: when privately informed agents strategically choose *when* to enter a sequential selection process, the entry order itself becomes an information channel for the hiring firm.

2. Literature Review

Our paper connects two bodies of literature: the classical secretary problem and the operations management literature on strategic agents.

2.1. The secretary problem.

The secretary problem studies optimal selection from a sequence of applicants who arrive in random order, and has generated a large literature in optimal stopping, probability, and operations research. Lindley (1961) formalized the classical version of this problem and developed the celebrated asymptotically optimal $1/e$ stopping rule, in the setting where (using the language of our present context) applicants' relative ranks are revealed upon review. A subsequent study by Gilbert and Mosteller (1966) analyzed an important variant of the problem in which the firm observes applicants' private quality scores upon review. Early developments of other variants of this problem are surveyed by Freeman (1983) and Ferguson (1989), while DeGroot (2005) places the problem within the broader theory of sequential decision making. Subsequent work has extended the classical problem along three major dimensions: the firm's objective, information structure, and action set, which we proceed to review below.

Objective. One strand of work modifies what the firm optimizes, replacing the goal of selecting the best candidate with objectives such as minimizing expected rank or maximizing expected payoff (Chow et al. 1964, Mucci 1973, Gianini and Samuels 1976, Lorenzen 1981). Related work studies assignment objectives, where multiple positions must be filled through sequential selection (Rose 1982, Tamaki 1984), and another line incorporates interview or sampling costs, so that the decision maker trades off the value

of additional search against the cost of delay (Bartoszyński and Govindarajulu 1978, Lorenzen 1981, Samuels 1985).

Information structure. A second strand modifies what the firm observes. Some papers replace ordinal ranks with cardinal scores (Gilbert and Mosteller 1966, Tamaki 1986, Correa et al. 2025); others consider noisy or biased rank evaluations (Salem and Gupta 2024); and still others relax knowledge of the horizon by allowing an uncertain number of applicants (Gianini-Pettitt 1979, Stewart 1981, Bruss 1984, Hill and Krengel 1991) or incorporate prior distributional information about candidate quality (Chun 1996, Zhang and Jaillet 2023, Fujii and Yoshida 2024).

Action set. A third strand introduces operational constraints on the firm’s actions. These include committee decision making (Alpern and Gal 2009, Alpern et al. 2010, Alpern and Baston 2017); recall, which allows previously rejected candidates to be reconsidered (Yang 1974, Karni and Schwartz 1977, Petruccielli 1981, Rose 1984b,a, Rocha 1993); multiple-choice variants, in which more than one candidate may be selected (Tamaki 1979, Preater 1994, Buchbinder et al. 2014, Fujii and Yoshida 2024, Salem and Gupta 2024); models in which candidates remain available for a limited window after arrival rather than requiring an immediate decision (Chow et al. 1964, Smith and Deely 1975, Kesselheim et al. 2024); and competition among firms searching the same applicant pool (Immorlica et al. 2006, Cownden and Steinsaltz 2014).

Across all three strands, one assumption remains: applicants are passive agents who arrive in a random order, with the firm as the sole strategic decision-maker. This is the assumption our paper relaxes. To our knowledge, only two prior papers have allowed applicants to behave strategically in a secretary-problem setting. Eriksson et al. (2007) study a two-sided secretary problem in which both sides make forward-looking decisions, and show that strategic incentives can fundamentally reshape the search process. The closest work to ours is an unpublished PhD dissertation by Truong (2013), which preserves the classical random interview order but allows candidates to choose how long to remain available after being interviewed, so that strategic behavior is tied to a randomly assigned position rather than to private quality information. Our paper departs from both: we study a setting in which all applicants strategically choose their arrival time as a function of private quality information, which transforms arrival order from an exogenous random sequence into an endogenous decision over applicant quality.

2.2. Strategic agents in operations management.

A separate body of work in operations management studies strategic customers and agents whose actions endogenously shape their interactions with firms or service systems. These papers share with ours an interest in strategic agents, but differ in the strategic choice under study. Kakade et al. (2013) develop an optimal dynamic mechanism design model with privately informed buyers. A substantial revenue-management literature studies how strategic customers time their purchases in response to dynamic prices or in anticipation of information revealed by earlier buyers (Su 2007, Caldentey and Vulcano 2007, Papanastasiou and Savva 2017, Chen and Farias 2018). A related literature studies strategic arrival timing in service systems, where customers choose when to join a queue in equilibrium, anticipating congestion (Honnappa and Jain 2015), or act on heterogeneous information about the system (Hu et al. 2018). Abhishek et al. (2021) study an online resource-allocation problem in which strategic customers choose when to submit requests in response to dynamically posted prices. Strategic timing also arises in decentralized matching markets, where the timing and deadlines of a proposer’s offer shape the counterparty’s search for alternatives (Zorac and Tsetlin 2020). In these models, agents’ strategic actions are purchase, queueing, request-timing, or offer-timing decisions driven by prices, congestion, information, or time preferences. These are distinct from what we study in this work where agents’ (i.e., applicants’) strategic actions determine when they would be evaluated as a function of their private quality, in response to a selection policy of the firm; it is this combination that renders the arrival order itself informative. A common conclusion across much of this literature is that strategic customer behavior is often detrimental to the firm, reflecting a misalignment of objectives. Our setting points in a different direction: the firm’s expected payoff can increase when applicants behave strategically, since arrival timing endogenously reveals information about private quality.

3. Model

3.1. Overview

Consider a firm (it) that seeks to hire a single secretary (they) for a single job position. There are $N + 1$ applicants who compete for the role, indexed by the set $\mathcal{N} := \{1, \dots, N + 1\}$, where the parameter $N \in \mathbb{N}$ is known to all players. We let $\mathbf{Q} = (Q_1, \dots, Q_{N+1})$ represent the vector of the applicants’ *quality scores* and $\mathbf{U} = (U_1, \dots, U_{N+1})$ the vector of *tie-breaking labels* (we discuss their role in §3.3). Each applicant $i \in \mathcal{N}$ observes only

their own quality score Q_i , and in particular, observes neither the qualities of the other applicants nor any tie-breaking label.¹

Similar to the classical secretary problem, we will assume that the firm interviews these applicants sequentially. However, our key departure from that model and other models in the extant literature is our assumption that the applicants are *strategic agents*. In our model of strategic agency, we will assume that each applicant aims to maximize their individual expected utility by choosing a finite time of submitting their application (normalized to $\mathcal{T} := [0, 1]$ without loss of generality), which in turn will endogenously determine their position in the sequence of interviews in a manner that we make precise in §3.3. One complication this brings is that we now have to carefully distinguish between each applicant's *index* (which is known *a priori*) and their *position* (which is unknown *a priori* because it is affected by their decisions). To improve clarity, we will adopt different linguistic conventions in each case. Specifically, we will use the phrase “applicant i ” to refer to the applicant with index i and the phrase “the j th applicant” to refer to the applicant who is in the position j of the interview sequence.

For each $i \in \mathcal{N}$, we denote applicant i 's decision (i.e., their application submission time) by t_i , the strategy profile of all applicants by the vector $\mathbf{t} = (t_1, \dots, t_{N+1})$, and the firm's decision with $\boldsymbol{\pi}$. In this baseline version of our model, we model the interaction between the firm and the $N + 1$ applicants as a Stackelberg game with the firm as a leader and the applicants as followers, and where all players use pure strategies. The sequence of events is as follows, and illustrated in Figure 1.

1. The firm chooses and commits to a hiring strategy $\boldsymbol{\pi}$ in anticipation of applicants' best responses.
2. The applicants select their arrival strategies $(t_1, \dots, t_{N+1})$ in response to the firm's choice of $\boldsymbol{\pi}$.
3. Applicants' private quality scores and their arrival times are realized.
4. The arrival times are mapped into a sequence of applicants' positions.
5. The firm implements its hiring strategy $\boldsymbol{\pi}$ over the realized sequence, and all utilities are realized.

¹Formally, we construct $\mathbf{Q}$ and $\mathbf{U}$ as random vectors defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P}) := ([0, 1]^{2(N+1)}, \mathcal{L}([0, 1]^{2(N+1)}), \lambda^{2(N+1)})$, where $\lambda^{2(N+1)}$ denotes the $2(N + 1)$ -dimensional Lebesgue measure on $[0, 1]^{2(N+1)}$, and $\mathcal{L}([0, 1]^{2(N+1)})$ denotes the Lebesgue σ -algebra (i.e., the completion of the Borel σ -algebra $\mathcal{B}([0, 1]^{2(N+1)})$ under $\lambda^{2(N+1)}$). We represent a generic sample point in this space as $\omega = (\mathbf{q}, \mathbf{u})$ with $\mathbf{q}, \mathbf{u} \in [0, 1]^{N+1}$, and define, for each $i \in \mathcal{N}$, the coordinate projections $Q_i(\omega) := q_i, U_i(\omega) := u_i$. Under $\mathbb{P}$, the random variables $\{Q_i, U_i\}_{i \in \mathcal{N}}$ are mutually independent and uniformly distributed on $[0, 1]$.



Figure 1 Sequence of events for the Stackelberg game between the firm (top) and applicants (bottom).

The Stackelberg assumption effectively means that the applicants can perfectly infer or observe the firm's committed hiring strategy. This is clearly an abstraction of reality, and our goal for this modeling assumption is simply to highlight the underlying incentives and strategic forces. In §6, we investigate a model variant in which the firm does not have this ability to commit, and discuss the differences in findings.

3.2. The firm's decisions

In this paper, we consider two distinct models of information that can be revealed by an interview: (1) a *score-based* information model where each interview reveals the exact realization of the applicant's quality score, and (2) a *rank-based* information model where each interview only reveals the applicant's rank relative to all previously interviewed applicants. Classic (i.e., non-strategic) versions of these problems were studied by Gilbert and Mosteller (1966) and Lindley (1961) respectively. As in these classical models, after each interview, we assume that the firm decides to hire or reject that applicant, and this decision is irrevocable in the sense that the firm cannot recall previously rejected applicants. The firm's objective is to maximize its probability of hiring the best applicant, receiving a unit of utility if it does so, and zero utility otherwise. To avoid unnecessary technical complications from handling corner cases, throughout, we assume that the firm hires the $(N + 1)$ th applicant if all previous applicants are rejected. This assumption is based on the rationale that hiring a suboptimal candidate is preferable to leaving the position unfilled.

With a mild abuse of notation, we denote a pure ("hiring") strategy of the firm in either information model by $\pi := (\pi_1, \dots, \pi_{N+1})$, where $\pi_j = 1$ means that the firm hires the j th applicant and $\pi_j = 0$ means that the firm rejects the j th applicant. The assumption from the previous paragraph translates to the requirement that $\pi_{N+1} \equiv 1$. It turns out that equipping applicants with agency severely complicates the firm's problem because at any

given decision point, the firm must now keep track of the information revealed by *all past applicants*. This is in contrast to classical versions of the secretary problem where it is sufficient to solely keep track of the information revealed by the *most recent applicant*.

Deferring a more in-depth discussion of this issue to §3.6, at present it suffices to note that the firm's strategies after interviewing the j th applicant have to depend on the full vector of scores (resp. ranks) observed up to that applicant. Denoting the strategy spaces of the firm in the score-based and rank-based information models by Π^S and Π^R respectively, we have

$$\Pi^S := \{(\pi_1, \dots, \pi_N, 1) \mid \pi_j : [0, 1]^j \rightarrow \{0, 1\}, \forall j = 1, \dots, N\}, \quad (1)$$

and

$$\Pi^R := \{(\pi_1, \dots, \pi_N, 1) \mid \pi_j : \mathcal{I}_j \rightarrow \{0, 1\}, \forall j = 1, \dots, N\}, \quad (2)$$

where we define $\mathcal{I}_j$ as the permutation group of $\{1, \dots, j\}$, i.e., the set of all bijections from $\{1, 2, \dots, j\}$ to itself.

3.3. Applicants' decisions and sequencing

Consider a generic applicant with index i , whose sole objective is to secure a position at the firm. They gain a unit of utility if they are hired, and zero otherwise. To maximize their expected utility (equivalently, their probability of being hired), they strategically select their application time t_i based on their observed quality score q and the firm's chosen strategy π . Formally, a pure arrival strategy for applicant i is a function $t_i : [0, 1] \times \Pi \rightarrow \mathcal{T}$, with $\Pi = \Pi^S$ in the score-based information model and $\Pi = \Pi^R$ in the rank-based information model. The applicants' arrival times determine the interview order as follows: applicants are interviewed in ascending order of t_i , with ties broken in descending order of the labels U_i introduced in §3.1. Since the U_i 's are i.i.d. uniform random variables independent of $\mathbf{Q}$, whenever a group of applicants chooses the same arrival time, every ordering within that group is equally likely. Figure 2 illustrates this mapping process. As illustrated, each applicant's position in the final interview sequence is determined by the actions of the entire pool of applicants, and after any potential randomized tie-breaking has occurred.

To represent the relationship between applicants' indices and their positions symbolically, consider a given strategy profile $\mathbf{t} = (t_1, \dots, t_{N+1})$. We define $\sigma_{\mathbf{t}} : \mathcal{N} \rightarrow \mathcal{N}$ as the

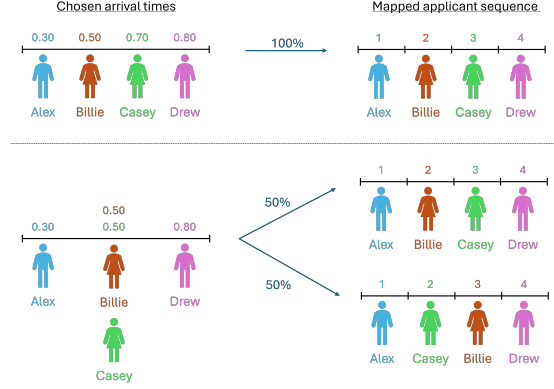


Figure 2 Mapping between applicants' chosen arrival times (left) and their sequence (right). Mapping is deterministic when all chosen times are distinct (top) and ties are broken uniformly at random (bottom).

mapping from an applicant's position to their index. Specifically, $\sigma_t(j)$ denotes the index of the j th applicant. For brevity, we suppress the dependence on t and write $\sigma(j)$ when it is obvious from context which t is used.

3.4. Players' objectives

Consider the firm after interviewing the j th applicant. In the score-based information model, the information available to the firm is the score vector of all applicants reviewed so far, denoted by $\mathbf{Q}_j := (Q_{\sigma(1)}, \dots, Q_{\sigma(j)})$. In the rank-based information model, we let $R_{k,j} = R_{k,j}^t$ denote the rank of the secretary in the k th position, relative to the first j secretaries, suppressing the explicit dependence on t when this is obvious from context. The relative ranks $R_{k,j}$ can be constructed from model primitives as $R_{k,j} := \sum_{k'=1}^j \mathbb{1}\{Q_{\sigma(k')} \geq Q_{\sigma(k)}\}$.

For example, $R_{k,j} = 1$ means that the k th applicant has the *highest* quality score among the first j applicants. In this second model of information, after reviewing the j th applicant, the information available to the firm is the vector $\mathbf{R}_j := (R_{1,j}, \dots, R_{j,j})$.

In either model, we will let $j_\pi^\dagger$ denote the position of the applicant who is hired, which we define as $j_\pi^\dagger := \min\{j \in \mathcal{N} : \pi_j = 1\}$, and note that this is well-defined since $\{j \in \mathcal{N} : \pi_j = 1\}$ is always non-empty by assumption. From this, we can write applicant i 's expected payoff when their score is q_i as the probability that they are selected by the firm, conditional on the realization of their score, i.e., $\mathbb{P}(\sigma_t(j_\pi^\dagger) = i | Q_i = q_i)$.

Finally, by writing $R_k = R_k^t := R_{k,N+1}^t$ to denote the *global rank* of the applicant in position k , that is, the rank of the k th applicant among all the $N+1$ applicants, the rank of the firm's hired applicant is $R_{j_\pi^\dagger}$, and the firm's objective is to maximize $\mathbb{P}(R_{j_\pi^\dagger} = 1)$.

Having specified the objectives of each player, we are ready to state a preliminary lemma which will help to simplify our subsequent arguments. This result is rather intuitive and straightforward, and we omit its proof for brevity.

LEMMA 1. *In the score-based information model, the firm achieves no less expected utility by restricting its space of policies to $\Pi_1^S \subseteq \Pi^S$, defined as*

$$\Pi_1^S := \left\{ \boldsymbol{\pi} \in \Pi^S : \pi_j(\mathbf{Q}_j) = 0 \quad \text{if } Q_{\sigma(j)} < \max_{k=1, \dots, j-1} Q_{\sigma(k)}, \quad \forall j = 2, \dots, N \right\}. \quad (3)$$

Similarly, in the rank-based information model, the firm achieves no less expected utility by restricting its space of policies to $\Pi_1^R \subseteq \Pi^R$, defined as

$$\Pi_1^R := \left\{ \boldsymbol{\pi} \in \Pi^R : \pi_j(\mathbf{R}_j) = 0 \quad \text{if } R_{j,j} \neq 1, \quad \forall j = 2, \dots, N \right\}. \quad (4)$$

This follows immediately from the observation that an applicant who is not the current best cannot be the globally best, making rejection without loss of generality. In light of Lemma 1, throughout the remainder of the paper we restrict the firm's strategy space to Π_1^S and Π_1^R in the score-based and rank-based information models, respectively.

3.5. Sequence of events and definition of equilibrium

In the analysis that follows, we focus on Stackelberg equilibria in which the firm commits to a stopping policy and applicants respond with a symmetric pure-strategy Bayesian Nash equilibrium (BNE). We refer to such an outcome as a symmetric BNE and define it formally below.

DEFINITION 1. Let $\Pi \in \{\Pi_1^S, \Pi_1^R\}$. A strategy profile $(\boldsymbol{\pi}^*, t^*)$ constitutes a *Stackelberg equilibrium with symmetric Bayesian Nash continuation equilibria* if the arrival strategy $t^* : [0, 1] \times \Pi \rightarrow \mathcal{T}$ satisfies the following: for every $\boldsymbol{\pi} \in \Pi$, there exists a set $\mathcal{Q}_\pi \subseteq [0, 1]$ with $\lambda(\mathcal{Q}_\pi) = 1$ such that

$$t^*(q, \boldsymbol{\pi}) \in \arg \max_{t \in \mathcal{T}} \mathbb{P}(\sigma_{(t, t^*(Q_2, \boldsymbol{\pi}), \dots, t^*(Q_{N+1}, \boldsymbol{\pi}))}(j_\pi^\dagger) = 1 \mid Q_1 = q), \quad \forall q \in \mathcal{Q}_\pi, \quad (5)$$

and the firm's committed strategy $\boldsymbol{\pi}^*$ satisfies

$$\boldsymbol{\pi}^* \in \arg \max_{\boldsymbol{\pi} \in \Pi} \mathbb{P}\left(R_{j_\pi^\dagger}^{(t^*(Q_1, \boldsymbol{\pi}), \dots, t^*(Q_{N+1}, \boldsymbol{\pi}))} = 1\right), \quad (6)$$

where $\Pi = \Pi_1^S$ in the score-based information model and $\Pi = \Pi_1^R$ in the rank-based information model.

Equation (5) represents a generic applicant's objective: choose an application time that maximizes their probability of being hired, based on any privately observed quality scores and a given strategy by the firm, and in response to the equilibrium actions of all other applicants. Since applicants' quality scores are continuously distributed on $[0, 1]$, we impose the best-response condition only almost everywhere; deviations on a measure-zero set of scores do not affect equilibrium outcomes. Therefore, throughout this paper, whenever we describe t^* as a *unique* equilibrium response, it should be interpreted as allowing for such zero-measure deviations. On the other hand, equation (6) means that the firm aims to optimize over its strategy space, Π_1^S or Π_1^R , in the score-based and rank-based information models respectively, to maximize the probability of hiring the global best applicant in the pool, when the applicants have symmetrically adopted their equilibrium strategies.

3.6. Comparison to the classical secretary problem

In the classical secretary problem, applicants arrive uniformly at random, so the firm can base its hiring decision on the rank or score of the *currently observed* applicant alone, together with whether that applicant is the best reviewed so far; it need not “remember” the full history. In our strategic version, this is no longer true: in general, the firm's optimal decision depends on the full history of past scores or ranks.

To illustrate, consider a toy example with $N + 1 = 4$ applicants in which the firm observes only rank-based information, and suppose, purely for illustration, that for any π , the applicants adopt the arrival strategy

$$t^*(q, \pi) = \begin{cases} 2q & \text{if } 0 \leq q \leq 0.5, \\ 1 & \text{if } 0.5 < q \leq 1, \end{cases}$$

plotted in Figure 3. Under t^* , applicants with scores below 0.5 arrive strictly in ascending order of their scores, while applicants with scores above 0.5 pool at the common time $t = 1$ and, by our tie-breaking rules, are sequenced uniformly at random. Consider the firm's problem at position 3, and compare two histories that share the property that the third arrival is the best among the first three: $H_{213} := \{Q_{\sigma(2)} < Q_{\sigma(1)} < Q_{\sigma(3)}\}$ and $H_{123} := \{Q_{\sigma(1)} < Q_{\sigma(2)} < Q_{\sigma(3)}\}$. On either history, the firm faces a binary choice at position 3: accept the third arrival, or reject them, in which case the fourth applicant is hired by

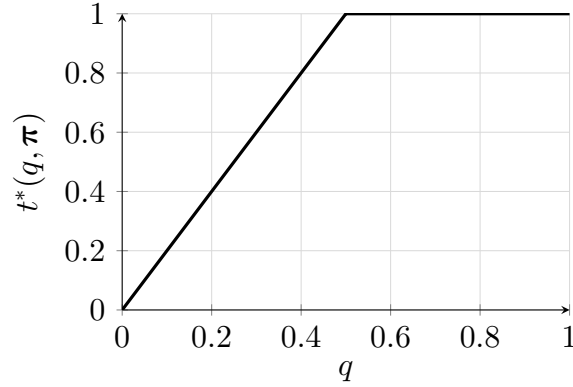


Figure 3 Illustrative example of $t^*(q, \pi)$

default. Because the third arrival is the best among the first three on both histories, accepting succeeds in hiring the global best if and only if $Q_{\sigma(3)} > Q_{\sigma(4)}$, while rejecting succeeds if and only if $Q_{\sigma(4)} > Q_{\sigma(3)}$. The firm's decision therefore turns on which of these two events is more likely, conditional on the observed history.

On H_{213} , the first arrival's score exceeds the second's, which under t^* requires $Q_{\sigma(2)} > 0.5$; since the second arrival is the lowest-scoring of the first three on H_{213} , all of the first three scores exceed 0.5. The fourth score must exceed 0.5 as well, since otherwise that applicant would arrive strictly before $t = 1$ and could not occupy the final position. Hence H_{213} arises only when all four applicants pool at $t = 1$ and are ordered uniformly at random, so that

$$\mathbb{P}(Q_{\sigma(4)} > Q_{\sigma(3)} \mid H_{213}) = \frac{\mathbb{P}(Q_{\sigma(2)} < Q_{\sigma(1)} < Q_{\sigma(3)} < Q_{\sigma(4)})}{\mathbb{P}(Q_{\sigma(2)} < Q_{\sigma(1)} < Q_{\sigma(3)})} = \frac{0.5^4 \cdot \frac{1}{4!}}{0.5^4 \cdot \frac{4}{4!}} = \frac{1}{4}.$$

In the numerator, the event requires the four pooled applicants to realize the unique ordering $Q_{\sigma(2)} < Q_{\sigma(1)} < Q_{\sigma(3)} < Q_{\sigma(4)}$, which occurs with probability $1/4!$; in the denominator, the pattern H_{213} fixes the relative order of the first three arrivals but leaves the fourth applicant's rank free, so exactly 4 of the $4!$ equally likely orderings are consistent with it. Since the fourth applicant is more likely to be outscored by the third, the firm strictly prefers to *accept* the third arrival.

On H_{123} , by contrast, some of the first three applicants may have scores below 0.5, and the calculation must account for every possible split of the pool. Let k denote the number of applicants whose scores lie in $(0.5, 1]$. Conditional on k , the $4 - k$ lower-scored

applicants arrive first in increasing order, while the k higher-scored applicants pool at $t = 1$ and are uniformly ordered among themselves. By the law of total probability,

$$\mathbb{P}(Q_{\sigma(4)} > Q_{\sigma(3)} \mid H_{123}) = \frac{\mathbb{P}(Q_{\sigma(1)} < Q_{\sigma(2)} < Q_{\sigma(3)} < Q_{\sigma(4)})}{\mathbb{P}(Q_{\sigma(1)} < Q_{\sigma(2)} < Q_{\sigma(3)})} = \frac{\sum_{k=0}^4 \binom{4}{k} 0.5^4 \frac{1}{k!}}{\sum_{k=0}^4 \binom{4}{k} 0.5^4 \frac{1}{\max\{k-1, 0\}!}} = \frac{209}{316} > \frac{1}{4}.$$

To see the two factorial terms, condition on exactly k applicants having scores above 0.5. The numerator event requires all k pooled applicants to appear in increasing-score order, which occurs with probability $1/k!$. For the denominator event, when $k \geq 1$, one pooled applicant occupies position 4, so only the remaining $k - 1$ pooled applicants must appear in increasing-score order among the first three arrivals; when $k = 0$, there are no pooled applicants and the event occurs automatically. This gives the factor $1/\max\{k - 1, 0\}!$. Since, conditional on H_{123} , the fourth applicant is more likely than the third to have the higher score, the firm strictly prefers to *reject* the third arrival and hire the fourth applicant by default.

Thus, two histories sharing the same relative rank of the current applicant (in both, the third arrival is the best observed so far) lead to opposite optimal actions. It is therefore generally insufficient for the firm to track only the most recent applicant's relative rank: its optimal decision may depend on the entire sequence of previously observed ranks, so the relevant state space grows with the history, and the firm's problem becomes a genuinely history-dependent dynamic program, in stark contrast to the classical secretary problem.

4. Illustrative case: Score-based information

In this section, we analyze the simplest special case of our model: two applicants (i.e., $N = 1$) under score-based information. This case illustrates the strategic tension faced by applicants and develops intuition for the general analysis, while demonstrating that the strategic problem is conceptually distinct from the classical one.

Recall from §3.2 that $\pi_2 = 1$: the applicant in position 2 is always hired by default. At position 1, the firm observes only the first applicant's score, so every feasible policy is equivalently characterized by a subset $A \subseteq [0, 1]$, which we call an *immediate acceptance set*: the firm commits ex ante to accept the first applicant if and only if $Q_{\sigma(1)} \in A$.

4.1. Equilibrium of the subgame between applicants.

We argue that for any policy π with immediate acceptance set A ,

$$t^*(q, \pi) = \begin{cases} 0 & \text{for almost every } q \in A, \\ 1 & \text{for almost every } q \in A^c, \end{cases} \quad (7)$$

is an equilibrium of the sub-game among applicants; the proof that it is the unique equilibrium is deferred to Theorem 1 in §5. Intuitively, an applicant whose score lies in A prefers to arrive early to secure the first position, whereas an applicant whose score lies outside A prefers to arrive late in the hope of being the second arrival.

To derive this structure, we focus on applicant 1 and let t_2 denote an arbitrary strategy of applicant 2; one such strategy is illustrated in Figure 4 for $A = [\frac{1}{2}, 1]$. Define

$$W_0 := \{q \in [0, 1] : t_2(q, \pi) = 0\}, \quad (8)$$

the set of scores for which applicant 2 arrives at time 0, along with $a := \lambda(A \cap W_0)$ and $b := \lambda(A^c \cap W_0)$ (see Figure 4), and write $p_1(t) := \mathbb{P}(\sigma_{(t, t_2(Q_2, \pi))}(j_\pi^\dagger) = 1 \mid Q_1 = q_1)$ for applicant 1's acceptance probability as a function of their arrival time t .

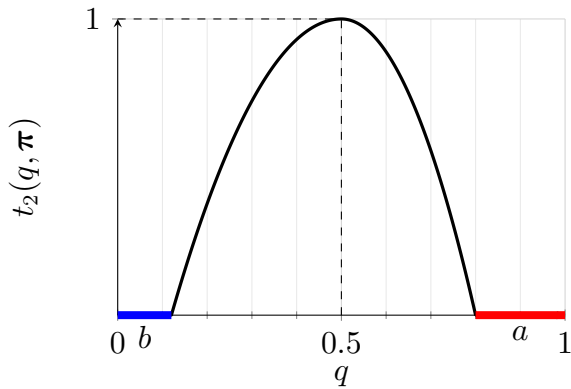


Figure 4 Illustrative example of a possible strategy $t_2(q, \pi)$ and the quantities a, b for the immediate acceptance set $A = [\frac{1}{2}, 1]$.

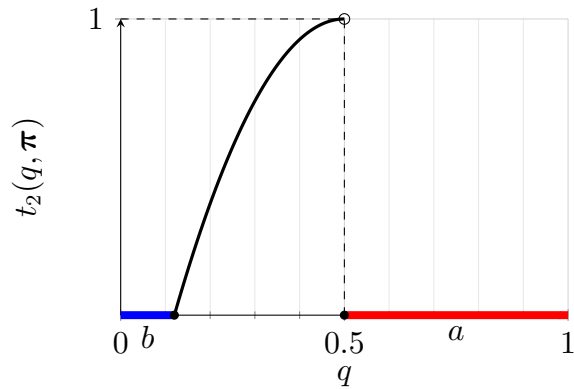


Figure 5 Illustrative example of a candidate strategy $t_2(q, \pi)$, with $A \subseteq W_0$, for the immediate acceptance set $A = [\frac{1}{2}, 1]$.

First, suppose $q_1 \in A$, so that applicant 1 is accepted with certainty whenever they occupy the first position. Partitioning on whether Q_2 lies in $A \cap W_0$, $A^c \cap W_0$, or W_0^c , we obtain

$$p_1(0) = \frac{1}{2}a + b + (1 - a - b) = 1 - \frac{a}{2}.$$

The first term arises because when both applicants arrive at time 0, tie-breaking places each in the first position with probability $\frac{1}{2}$; the second because if applicant 2 wins the tie, they are rejected ($Q_2 \in A^c$) and applicant 1 is accepted by default in position 2; and the third because applicant 1 is then the first arrival with certainty. By contrast, for any $t_1 > 0$, on the event $Q_2 \in A \cap W_0$ (probability a) applicant 2 arrives first and is accepted, so $p_1(t_1) \leq 1 - a \leq p_1(0)$. Hence applicant 1's best response is $t_1 = 0$. This argument did not use any structure of t_2 ; by symmetry, applicant 2's best response likewise satisfies $t_2(q_2, \pi) = 0$ for almost every $q_2 \in A$, so we may henceforth restrict attention to candidate strategies with $A \subseteq W_0$, as illustrated in Figure 5.

Next, suppose $q_1 \in A^c$, and fix any t_2 with $A \subseteq W_0$. Applicant 1 is rejected in the first position, so they are accepted only if applicant 2 occupies position 1 and is rejected there, i.e., only if $Q_2 \in A^c$. If $t_1 = 0$, this requires winning the tie against $Q_2 \in A^c \cap W_0$, so $p_1(0) = \frac{1}{2}b$. If $t_1 > 0$, then $Q_2 \in A^c \cap W_0$ guarantees acceptance (applicant 2 arrives first and is rejected), while $Q_2 \in W_0^c \subseteq A^c$ yields acceptance exactly when applicant 1 is sequenced second:

$$p_1(t_1) = b + (1 - a - b) \left[\mathbb{P}(t_1 > t_2(Q_2, \pi) \mid Q_2 \in W_0^c) + \frac{1}{2} \mathbb{P}(t_1 = t_2(Q_2, \pi) \mid Q_2 \in W_0^c) \right].$$

Since $p_1(t_1) \geq b = 2p_1(0)$ and $p_1(t_1)$ is weakly increasing in t_1 , applicant 1's best response is $t_1 = 1$, and (7) is applicant 1's best response to any t_2 with $A \subseteq W_0$. In fact, (7) is a (weakly) dominant strategy: for *arbitrary* t_2 , an applicant with score in A^c can be accepted only when $Q_2 \in A^c$ and they occupy position 2, an event whose probability is weakly increasing in t_1 and thus maximized at $t_1 = 1$, while the case $q_1 \in A$ was already handled for arbitrary t_2 above. By symmetry, the same holds for applicant 2.

4.2. The firm's optimal policy.

Anticipating the applicants' equilibrium response (7) to any π , the firm's problem reduces to choosing an optimal immediate acceptance set A . Any realization (Q_1, Q_2) falls into one of three cases: (a) both scores in A ; (b) both in A^c ; or (c) exactly one score in each. In cases (a) and (b) both applicants arrive simultaneously, the interview sequence is random, and the firm hires the better candidate with probability $1/2$ regardless of A . In case (c), the firm accepts the applicant whose score is in A , and succeeds whenever that score exceeds the other. Writing $p_{best}(A) = \mathbb{P}\left(R_{j_\pi}^{(t^*, t^*)} = 1\right)$, we obtain

$$p_{best}(A) = \frac{1}{2}\lambda(A)^2 + \frac{1}{2}[1 - \lambda(A)]^2 + 2\lambda(A)[1 - \lambda(A)] \mathbb{P}(Q_1 > Q_2 \mid Q_1 \in A, Q_2 \in A^c).$$

The firm can restrict attention to upper intervals: for any A , the set $\tilde{A} := [1 - \lambda(A), 1]$ satisfies $\lambda(\tilde{A}) = \lambda(A)$ and $\mathbb{P}(Q_1 > Q_2 \mid Q_1 \in \tilde{A}, Q_2 \in \tilde{A}^c) = 1$, so $p_{best}(A) \leq p_{best}(\tilde{A})$. For $A = [\zeta, 1]$, $p_{best}(A) = \frac{1}{2}(1 - \zeta)^2 + \frac{1}{2}\zeta^2 + 2\zeta(1 - \zeta)$, which is maximized at $\zeta = 1/2$ with value $3/4$.

In summary, when $N = 1$ the firm's optimal policy is a threshold rule: accept the first arrival if and only if its score exceeds 0.5. In equilibrium, a separating structure emerges among the applicants: higher-quality applicants (scores above $1/2$) sort themselves by arriving at $t = 0$, and lower-quality applicants by arriving at $t = 1$. Disappointingly for the firm, however, it cannot exploit this sorting: its equilibrium probability of hiring the best applicant is 0.75, identical to the classic secretary problem with score-based information and two applicants (Gilbert and Mosteller 1966). In the next section, we investigate whether this equivalence persists for general N , under both score-based and rank-based information.

5. Main Results

As established before, the strategic interaction among applicants substantially complicates the firm's optimization problem. We now analyze the equilibrium for both the score-based and rank-based information models. Because the analyses for both models follow similar conceptual arguments, we present them simultaneously, at the mild expense of slightly more notation and case-handling.

5.1. The Crucial Role of π_1 .

As a first step toward analyzing this game, we examine the firm's policy at position 1, π_1 , and show that its structure has a strong influence on the form of the applicants' best-response arrival strategy $t^*(\cdot, \pi)$. The key observation is that π_1 depends solely on the information revealed by the first applicant. In the score-based model, the firm observes only the first applicant's score, and therefore π_1 can be equivalently described by an *immediate acceptance set* A as follows:

$$\pi_1(Q_{\sigma(1)}) = \mathbb{1}\{Q_{\sigma(1)} \in A\}. \quad (9)$$

In the rank-based model, the firm gains no information from the first applicant and must therefore commit ex ante to $\pi_1 \in \{0, 1\}$. The immediate acceptance set A remains a valid

representation in this case, with the restriction that $A = [0, 1]$, corresponding to $\pi_1 \equiv 1$, or $A = \emptyset$, corresponding to $\pi_1 \equiv 0$.

We next show that the firm's choice of A plays a significant role in shaping the applicants' best-response arrival strategy $t^*(\cdot, \pi)$. To facilitate the subsequent discussion, we define the level set of quality scores for which applicants choose arrival time $t \geq 0$ as follows:

$$W_t := \{q \in [0, 1] : t^*(q, \pi) = t\}. \quad (10)$$

PROPOSITION 1. *For any $\pi \in \Pi_1^S \cup \Pi_1^R$ with a corresponding immediate acceptance set A , suppose that t^* is an equilibrium strategy implemented by the applicants. Let W_0 be as defined in (10). It is necessary that $\lambda(W_0 \cap A^c) = 0$.*

Proposition 1 states that under any firm policy, almost every applicant whose score lies outside A chooses a strictly positive arrival time $t > 0$. Intuitively, since the firm rejects the first applicant whenever their score lies in A^c , such an applicant seeks to avoid arriving first and therefore has an incentive to delay. Building on this, we now establish a converse result characterizing equilibrium behavior for applicants whose scores lie inside A .

PROPOSITION 2. *For any $\pi \in \Pi_1^S \cup \Pi_1^R$ with a corresponding immediate acceptance set A , suppose that t^* is an equilibrium strategy implemented by the applicants. Let W_0 be as defined in (10). It is necessary that $\lambda(W_0^c \cap A) = 0$.*

The intuition behind Proposition 2 is straightforward as well: since the firm accepts any applicant in the first position whose score falls in A , such applicants are incentivized to arrive as early as possible, that is, at time 0, to secure the first position before being preempted by another applicant.

Together, Propositions 1 and 2 yield a sharp characterization of the set of applicants who choose to arrive at time 0 in equilibrium. We state this in the following corollary, whose proof follows directly from the preceding propositions and is omitted for brevity.

COROLLARY 1. *For any $\pi \in \Pi_1^S \cup \Pi_1^R$ with a corresponding immediate acceptance set A , in any equilibrium strategy implemented by the applicants, it is necessary that the set of quality scores for which applicants select the arrival time of $t = 0$ is almost surely equal to the immediate acceptance set A . That is, $\lambda(W_0 \cap A) = \lambda(W_0) = \lambda(A)$.*

The above analysis highlights the crucial role played by π_1 , and equivalently by the acceptance set A , in shaping applicants' equilibrium behavior. In particular, even without imposing any structural assumptions on the firm's policies π_2 through π_N , the policy at position 1 alone suffices to characterize the coarse structure of the applicants' best-response arrival strategy t^* : which score realizations lead to arrival at exactly time 0, and which lead to arrival strictly after time 0 almost surely.

One useful consequence of Corollary 1 is that it allows us to simplify the firm's objective. For each $k \in \mathcal{N}$, let E_k denote the event that exactly k applicants have scores in A , so that

$$\mathbb{P}(E_k) = \binom{N+1}{k} \lambda(A)^k [1 - \lambda(A)]^{N+1-k}. \quad (11)$$

Applying the law of total probability, the firm's objective decomposes as

$$\mathbb{P}(R_{j_\pi^\dagger} = 1) = \mathbb{P}(R_{j_\pi^\dagger} = 1 \mid E_0) \mathbb{P}(E_0) + \sum_{k=1}^{N+1} \mathbb{P}(R_1 = 1 \mid E_k) \mathbb{P}(E_k). \quad (12)$$

For $k \geq 1$, on event E_k , at least one applicant has a score in A . By Corollary 1, such applicants almost surely arrive at time 0, so the firm almost surely hires the first applicant, i.e., $\mathbb{P}(j_\pi^\dagger = 1 \mid E_k) = 1$, which yields the expression on the right-hand side of (12).

5.2. Equilibrium of the subgame between applicants

Having clarified the critical role of the firm's first-position policy and the immediate acceptance set in shaping applicants' equilibrium strategy structure, we now fully characterize applicants' best responses for a given $\pi \in \Pi_1^S \cup \Pi_1^R$ committed to by the firm. As a preliminary step, the following lemma shows that no best response $t^*(\cdot, \pi)$ can be strictly increasing almost everywhere on A^c .²

LEMMA 2. *For any $\pi \in \Pi_1^S \cup \Pi_1^R$ with a corresponding immediate acceptance set A satisfying $\lambda(A^c) > 0$, suppose that t^* is an equilibrium strategy implemented by the applicants. Then, $t^*(\cdot, \pi)$ cannot be strictly increasing almost everywhere on A^c .*

The intuition behind Lemma 2 is as follows. An applicant with a score in A^c can be accepted only when *all* applicants have scores in A^c , since whenever at least one applicant has a score in A , that applicant is accepted in position 1, leaving no possibility

²We say that a function f is *strictly increasing almost everywhere* on a set $S \subseteq [0, 1]$ if $\lambda^2(\{(q, q') \in S \times S : q < q', f(q) \geq f(q')\}) = 0$, i.e., if the set of score pairs on which f is weakly inverted is λ^2 -null.

of acceptance for any applicant in A^c . It therefore suffices, from the perspective of any applicant with a score in A^c , to focus on the event that all $N + 1$ applicants have scores in A^c . Conditioning on this event, suppose the best response $t^*(\cdot, \pi)$ is strictly increasing almost everywhere on A^c . Two cases arise. In the first, the firm's strategy π reaches the $(N + 1)$ th applicant with positive probability. In this case, *low-score applicants* in A^c who arrive early under t^* — and are therefore likely to be the first arrival and be rejected — have a profitable deviation to the latest possible arrival time, where they can be hired in the last position with positive probability. In the second case, the firm's π reaches the $(N + 1)$ th applicant with zero probability. In this case, *high-score applicants* in A^c who arrive late under t^* risk ending up in the unreachable last position, and therefore have a profitable deviation to an earlier arrival time. In either case, strict monotonicity on A^c generates a profitable deviation for some applicants. We refer the reader to the proof of Lemma 2 for the formal argument.

We next establish a further structural property of applicants' best-response strategies.

LEMMA 3. *Let $\pi \in \Pi_1^S \cup \Pi_1^R$ and assume that $\lambda(A^c) > 0$. Letting $t^*(\cdot, \pi)$ be an equilibrium best response adopted by the applicants, it is necessary that*

$$\mathbb{P}(t^*(\max\{Q_1, \dots, Q_N\}, \pi) = \min\{t^*(Q_1, \pi), \dots, t^*(Q_N, \pi)\} | Q_1, \dots, Q_N \in A^c) > 0.$$

In words, Lemma 3 establishes that, conditional on all N applicants having scores in A^c , there is always a strictly positive probability that the highest-scoring applicant arrives earliest. One implication of this result, used in our subsequent analysis, is that the last-arriving applicant, the $(N + 1)$ th arrival, is accepted with strictly positive probability, which follows from combining this lemma with the structure of $\Pi_1^S \cup \Pi_1^R$ established in Lemma 1.

We are now ready to state the first key result of this paper: a sharp characterization of the symmetric equilibrium among applicants.

THEOREM 1. *For any firm policy $\pi \in \Pi_1^S \cup \Pi_1^R$ with corresponding immediate acceptance set A , there exists a unique symmetric equilibrium among the applicants, $t^*(\cdot, \pi)$, with the following structure:*

$$t^*(q, \pi) = \begin{cases} 0, & \text{for almost every } q \in A, \\ 1, & \text{for almost every } q \in A^c. \end{cases} \quad (13)$$

That is, in equilibrium, applicants with scores in A pool at time 0 almost surely, while applicants with scores outside A pool at time 1 almost surely.

Theorem 1 implies that, for any policy committed to by the firm, applicants will partially pool their arrivals into two groups: (i) those with scores in the firm’s immediate acceptance region A will almost surely choose to arrive at time 0; and (ii) those with scores outside of A will almost surely choose to arrive at time 1. For the score-based model in particular, this serves as a generalization of the equilibrium structure characterized in the case of two applicants (i.e., $N = 1$) that we discussed in §4. We note that for any given A , this equilibrium response of applicants is unique up to sets of zero measure.

Some of the intuition underlying this result has already been established by Corollary 1: applicants with scores in A arrive at time 0 to maximize their chances of occupying the first position and being accepted, while applicants with scores outside A delay their arrivals to minimize their risk of being rejected in the first position. What Theorem 1 adds is a complete characterization of the latter group’s best response, showing that these applicants *maximally* delay their arrival to the latest possible time, $t = 1$.

The intuition is as follows. If applicants in A^c did not choose a maximal delay, the applicant arriving earliest among them, call them “Alex”, would face pressure from two opposing forces. On the one hand, if any of Alex’s opponents has a score in A , Alex is rejected because that opponent is accepted in the first position. On the other hand, by being the earliest arrival among applicants in A^c , Alex has a high likelihood of occupying the first position when all opponents are also in A^c , and would consequently be rejected there as well. The preceding discussion is a qualitative argument and is not technically precise, the “earliest” arrival may not exist, and there may be multiple applicants tied for the earliest arrival time, and we refer the reader to the formal proof for the precise treatment of these issues.

5.3. The firm’s optimal policy

Theorem 1 confers substantial structure to the equilibrium of the overall game and, in particular, to the firm’s optimization problem. The firm’s choice of A is highly consequential in shaping applicants’ equilibrium responses, and we now exploit this structure to characterize the firm’s optimal policies. While the preceding analysis was presented jointly for both information models, we now find it convenient to treat each model separately.

5.3.1. Score-based information model. We first characterize the firm's optimal policy in the score-based information model.

THEOREM 2. *Suppose the firm observes score-based information about applicants. For $j = 1, \dots, N + 1$, let θ_j denote the optimal threshold at position j in the classical score-based secretary problem when there are $N + 1$ secretaries in total, as characterized by equation (3b-1) in Gilbert and Mosteller (1966). Then, there exists $L \in [0, 1]$ such that the firm's optimal policy $\pi^* = (\pi_1^*, \dots, \pi_N^*, 1)$ is given by $\pi_1^*(Q_{\sigma(1)}) = \mathbb{1}\{Q_{\sigma(1)} \geq L\}$ and*

$$\pi_j^*(Q_j) = \begin{cases} 1, & \text{if } Q_{\sigma(j)} \geq \max\{L\theta_j, Q_{\sigma(1)}, \dots, Q_{\sigma(j-1)}\}, \\ 0, & \text{otherwise,} \end{cases} \quad j = 2, \dots, N. \quad (14)$$

The firm's optimal policy in our strategic secretary problem shares the same qualitative structure as that of the classical score-based secretary problem, in that it is also characterized by a sequence of monotonically decreasing thresholds. Under π^* , the applicants' equilibrium best response from Theorem 1 simplifies to:

$$t^*(q, \pi^*) = \begin{cases} 0, & \text{for almost every } q \geq L \\ 1, & \text{for almost every } q < L. \end{cases} \quad (15)$$

An important consequence of Theorem 2 is that the firm's problem reduces to a one-dimensional optimization over $L \in [0, 1]$.

PROPOSITION 3. *There exists a unique optimal cutoff $L^* \in (0, 1)$ for the firm. It is obtained by maximizing the quasiconcave function $\Phi : [0, 1] \rightarrow \mathbb{R}_+$ defined by*

$$\Phi(L) = L^{N+1} \Gamma_N + \sum_{k=1}^{N+1} \frac{1}{k} \binom{N+1}{k} (1-L)^k L^{N+1-k},$$

where

$$\Gamma_N := \sum_{r=1}^N \left[\frac{1}{r(N+1)} + \frac{1}{N+1-r} \sum_{i=2}^r \left(\frac{\theta_i^r}{r} - \frac{\theta_i^{N+1}}{N+1} \right) - \frac{\theta_{r+1}^{N+1}}{N+1} \right].$$

Note that Γ_N follows directly from the winning probability formula (3c-1) of Gilbert and Mosteller (1966), evaluated at the threshold sequence $(1, \theta_2, \dots, \theta_{N+1})$ induced by the firm's optimal policy in Theorem 2, where the leading threshold of 1 encodes the rejection of the first arrival when no applicant's score exceeds L . Proposition 3 completes the characterization of the equilibrium of the overall game. In the left panel of Figure 6,

we illustrate this result by plotting the firm's success probability as a function of L for various values of N , from which the quasiconcave shape is evident. The right panel shows that the optimal L^* increases with N : as the applicant pool grows, the firm becomes more selective at the first position, reflecting the same trade-off present in the classical model of Gilbert and Mosteller (1966).

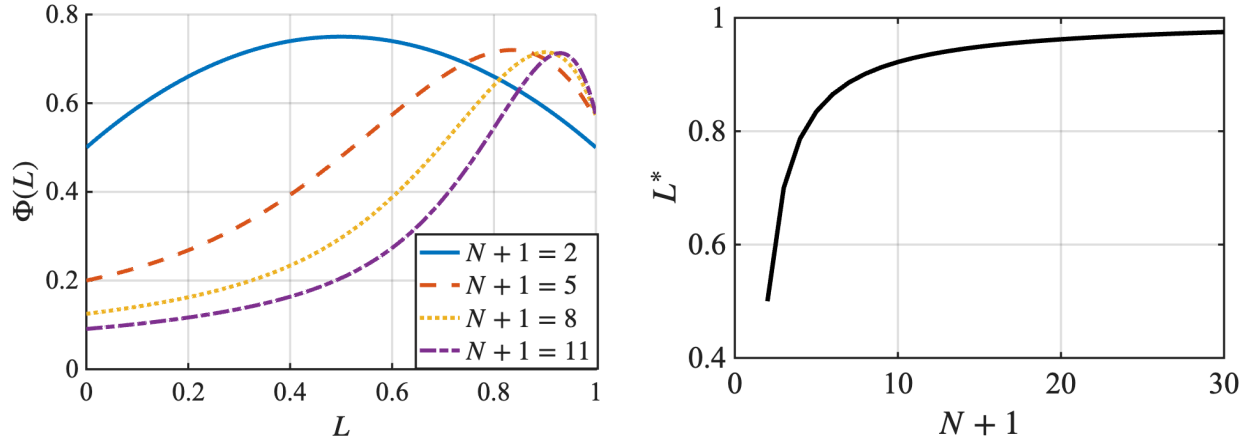


Figure 6 Left: The firm's success probability as a function of L . Right: The firm's optimal first-position threshold L^* as a function of $N+1$.

Figure 7 compares the optimal thresholds in our model with those in the classical model, across values of N for fixed positions (top row) and across positions for fixed values of N (bottom row). For $N \geq 2$, the optimal threshold L^* in our model at position 1 is higher than the corresponding classical threshold θ_1 (top left panel), while at positions 2 through N , the thresholds in our model are lower than their classical counterparts (top right panel), a direct consequence of Theorem 2 since $L^*\theta_j \leq \theta_j$ for all $j \geq 2$. The bottom row illustrates both patterns with a three-applicant and a ten-applicant example. The intuition behind these patterns is as follows. In equilibrium, all applicants with scores above the cutoff L^* arrive at time 0, and the firm hires one of them uniformly at random. Therefore, if there are k such applicants, the firm succeeds with probability $1/k$. A moderate cutoff thus invites a large cluster that dilutes the chance of hiring the best; to keep the cluster small and elite, the firm holds position 1 to a stricter standard than the classical threshold. If instead the first arrival is rejected and the firm reaches position 2, it learns that no applicant cleared the bar: it has entered a “market for lemons” in

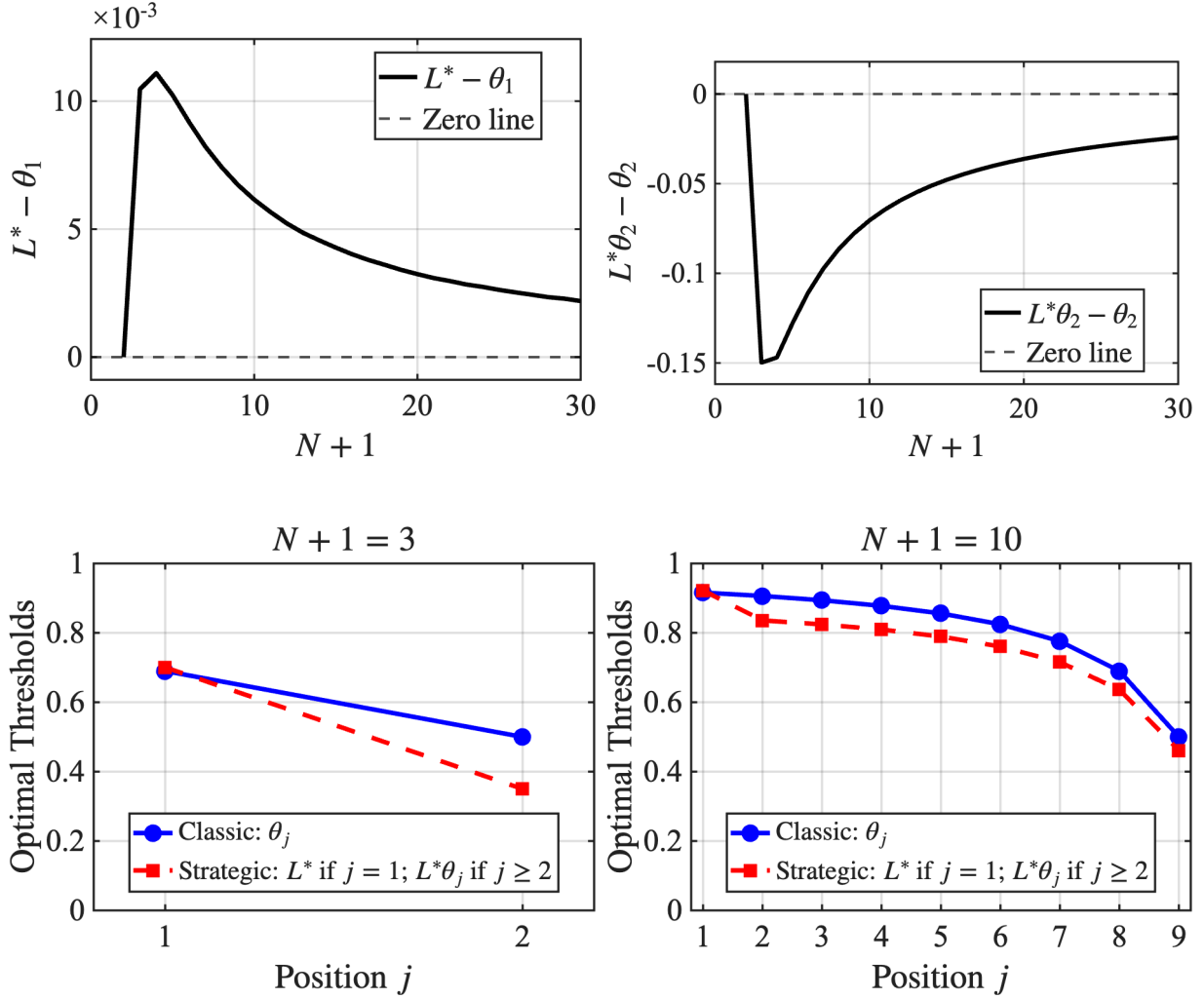


Figure 7 Optimal score thresholds in the strategic score-based model versus the classical score-based secretary problem. **Top row:** thresholds at position $j = 1$ (left) and position $j = 2$ (right), as functions of the total number of applicants $N + 1$. **Bottom row:** thresholds across positions j for $N + 1 = 3$ (left) and $N + 1 = 10$ (right) total applicants.

which all remaining scores lie below L^* , and it accordingly relaxes its acceptance criteria at every subsequent position.

By numerically evaluating the firm's optimized expected utility (i.e., its optimal probability of hiring the best applicant) for various values of N , we find that, similar to the classical model, the firm's expected utility decreases with N . More importantly, the firm's expected utility is always strictly higher in our model compared to that of the classical model when $N \geq 2$, even though they were equal for $N = 1$ (as shown in §4). Indeed, our numerical results suggest that, as $N \rightarrow \infty$, the success probability in our model approaches approximately 0.71, which is substantially higher than the analogous limit of

0.58 for the classical model (Figure 8). In our model, because applicants are strategic, they partially self-select into the times $t = 0$ and $t = 1$, with “better” applicants preferentially choosing the former. The firm thereby benefits from this self-selection, resulting in it having a higher expected utility.

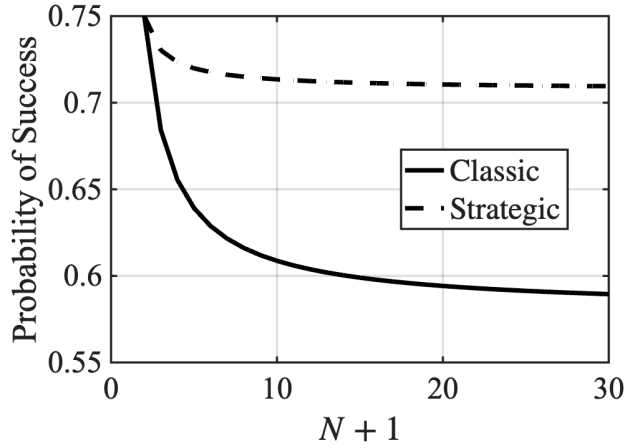


Figure 8 The firm’s optimized probability of hiring the best applicant in the strategic secretary problem (this paper) and the classical secretary problem (Gilbert and Mosteller 1966).

Finally, and perhaps most surprisingly, recall that we have shown in §3.6 that when the applicants are strategic agents, the firm must, *a priori*, consider all history-dependent policies in its optimization problem. Nevertheless, Theorem 2 implies that, after characterizing the structure of applicants’ equilibrium best responses and optimizing over the policies of the firm, the final optimal policy in every stage only depends on the current applicant’s score and whether or not that applicant is the best so far. We note that this generalizes the case of two applicants in §4, which corresponds to the case of $N = 1$ and $L = 1/2$.

Taken together, these results demonstrate that when applicants strategically choose their arrival times, a firm that sequentially observes applicants’ quality scores can achieve strictly better hiring outcomes than in the classical model with an exogenous arrival order. In equilibrium, applicants follow a two-step arrival strategy characterized by a jump at L , the optimal threshold set by the firm for the first arrival.

5.3.2. Rank-based information model. We now characterize the firm’s optimal policy under the rank-based information model, and ask whether the firm can similarly

benefit from strategic applicants when it observes only rank-based information. Perhaps surprisingly, the answer is negative, as stated in Theorem 3.

THEOREM 3. *Suppose that the firm has access only to rank-based information about applicants. Then the firm's optimal policy coincides with that of the classical secretary problem by Lindley (1961). In particular, there exists an optimal cutoff $r_N^* \in \{1, \dots, N\}$ such that*

$$\pi_j^*(\mathbf{R}_j) = 0 \quad \text{for } j = 1, \dots, r_N^* - 1,$$

and for $j = r_N^*, \dots, N$,

$$\pi_j^*(\mathbf{R}_j) = \begin{cases} 1, & \text{if } R_{j,j} = 1, \\ 0, & \text{otherwise.} \end{cases}$$

That is, the firm rejects the first $r_N^* - 1$ applicants and then accepts the first subsequent applicant who ranks best among all applicants observed so far. Moreover, $\lim_{N \rightarrow \infty} \frac{r_N^*}{N+1} = \frac{1}{e}$.

The main implication of Theorem 3 is that the firm's optimal policy coincides exactly with the classical secretary policy of Lindley (1961). The reasoning is as follows. By Theorem 1, in the rank-based information model the set A is either the entire unit interval or the empty set. In either case, applicants arrive in an effectively random order, with all $(N+1)!$ permutations equally likely after tie-breaking. Consequently, the firm faces precisely the classical secretary problem and achieves the same success probability as in the classical model.

6. Extension: When the Firm Cannot Commit

We now investigate a variant of our model in which the firm can no longer credibly commit to its hiring strategy. In this variant, the firm chooses its hiring strategy π and all $N+1$ applicants choose their arrival strategies $(t_1, \dots, t_{N+1})$ simultaneously. After these choices, the model proceeds identically to the base model in Section 5: applicants' private quality scores and arrival times are realized, the realized arrival times determine positions, and the firm implements its hiring strategy over the realized sequence. The sequence of events is illustrated in Figure 9.

We let Π^S and Π^R denote the firm's strategy spaces in the simultaneous-move model under score-based and rank-based information, respectively. The argument for Lemma 1 continues to apply, so we again restrict attention to $\Pi_1^S \cup \Pi_1^R$.



Figure 9 Sequence of events in the simultaneous-move game between the firm (top) and applicants (bottom).

In this variant, the interaction between the firm and applicants constitutes a simultaneous-move Bayesian game, and the relevant solution concept is the pure-strategy symmetric Bayesian Nash equilibrium. We formally define such equilibria below.

DEFINITION 2. Let $\Pi \in \{\Pi_1^S, \Pi_1^R\}$. A strategy profile (π^*, t^*) constitutes a *symmetric Bayesian Nash equilibrium (BNE)* of the simultaneous move game between the firm and applicants if there exists some set $\mathcal{Q} \subseteq [0, 1]$ with $\lambda(\mathcal{Q}) = 1$ such that the applicants' symmetric arrival strategy $t^* : [0, 1] \rightarrow \mathcal{T}$ satisfies

$$t^*(q) \in \arg \max_t \mathbb{P} \left(\sigma_{(t, t^*(Q_2), \dots, t^*(Q_{N+1}))}(j_{\pi^*}^\dagger) = 1 \mid Q_1 = q \right), \quad \forall q \in \mathcal{Q}, \quad (16)$$

given the firm's strategy π^* , and the firm's strategy π^* satisfies

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathbb{P} \left(R_{j_{\pi}^\dagger}^{(t^*(Q_1), \dots, t^*(Q_{N+1}))} = 1 \right), \quad (17)$$

given the applicants' strategy t^* .

The key difference from the base model is that the applicants' equilibrium strategy t^* is no longer a function of the firm's strategy π .

Despite this difference, several important elements of the base-case analysis carry over. Any feasible firm policy π has an associated immediate acceptance set $A \subseteq [0, 1]$, and Theorem 1, which characterizes the structure of a symmetric best response of applicants, remains valid in this setting. Specifically, if t^* forms part of a BNE in this variant, it is necessary that

$$t^*(q) = \begin{cases} 0, & \text{for almost every } q \in A, \\ 1, & \text{for almost every } q \notin A. \end{cases} \quad (18)$$

We now analyze the equilibria of this variant under each information model.

6.1. Score-based information model

Although (18) characterizes the applicants' best response t^* to a given firm policy π , for π to form part of a BNE it must itself be a best response to t^* . The following proposition shows that no such fixed point exists.

PROPOSITION 4. *Assume that the firm (a) has access to score-based information about applicants and (b) cannot commit to its hiring strategy. Then, there is no (π^*, t^*) that constitutes a symmetric pure-strategy BNE of the simultaneous move game between the firm and applicants.*

Proposition 4 points to an unfavorable implication for the firm. The issue is not merely that the firm may fail to improve its utility when applicants are strategic; under the score-based information model, no pure-strategy symmetric BNE exists at all.

The key idea underlying the proof is that, as a best response to (18), the firm always has an incentive to expand its immediate acceptance set A by adding the highest-scoring elements of A^c . In the base model, the firm's commitment power prevented such deviations. Without commitment, this incentive persists, preventing the formation of a pure-strategy symmetric BNE.

This result underscores the central role of the firm's commitment power when applicants are strategic. In the Stackelberg model, commitment enables the firm to achieve a higher payoff than in the classical model. Here, the absence of commitment power precludes the existence of a pure-strategy equilibrium altogether. More broadly, strategic behavior by applicants alone is not sufficient to improve hiring outcomes; for such interactions to be well-behaved, the firm must also possess sufficient commitment power.

6.2. Rank-based information model

As in the Stackelberg setting, the rank-based information model restricts A to either $[0, 1]$ or $\emptyset$, and by (18), applicants' best responses reduce to pooling at a common arrival time, so that every permutation of the interview sequence is equally likely after tie-breaking. The firm therefore faces exactly the classical rank-based problem of Lindley (1961), and the equilibria of the simultaneous-move game follow directly from the classical solution: when $N \geq 2$, the classical optimal policy is unique and rejects the first arrival, which in turn pins down the applicants' pooling time at 1; when $N = 1$, the firm is indifferent between accepting and rejecting the first arrival in the classical two-applicant problem,

and each of the two classically optimal policies supports its own pooling equilibrium. Verifying mutual best responses in each case is straightforward, so we omit the proof and summarize the result below.

PROPOSITION 5. *Assume that the firm (a) has access only to rank-based information about applicants and (b) cannot commit to its hiring strategy. Then, in any symmetric BNE of the simultaneous-move game between the firm and the applicants, all applicants pool at a common arrival time. When $N \geq 2$, the equilibrium is unique: all applicants choose $t^*(q) = 1$ for almost every q , and the firm's equilibrium policy coincides with the classical cutoff rule of Lindley (1961), as described in Theorem 3. When $N = 1$, mirroring the multiplicity of optimal policies in the two-applicant classical problem, there are exactly two symmetric BNE: the firm rejects the first arrival and all applicants choose $t^*(q) = 1$ for almost every q , or the firm accepts the first arrival and all applicants choose $t^*(q) = 0$ for almost every q .*

As in the Stackelberg setting, when the firm cannot separate applicants under any feasible policy, it faces exactly the classical secretary problem and achieves the same success probability as in the classical model. This holds for every equilibrium of the game: even where multiple equilibria arise, they are payoff-equivalent for the firm. The comparison thus highlights the value of information: when the firm observes only rank-based information, it cannot benefit from applicants' strategic behavior, regardless of whether it possesses commitment power.

7. Conclusion

Motivated by online labor platforms such as Upwork, where applicants to a single position may strategically control when they are reviewed by the firm, we study a feature that is absent in the classical version of the secretary problem and its extant variants: the endogeneity of applicants' arrival times. Our analysis addresses two related questions: how do applicants behave in equilibrium when arrival times are endogenous, and how does their strategic behavior affect the firm's probability of hiring the best applicant? We find that the answer to the latter question depends critically on two factors, namely, (a) whether the firm can commit to its chosen hiring strategy, and (b) whether the firm has access to rank-based or score-based information. We summarize these findings in Table 1.

		Information Accessibility	
		Score-based	Rank-based
Commitment Power	Yes	✓	×
	No	–	×

Table 1 Does the firm strictly benefit if applicants are strategic? ✓: Yes, ×: No, –: No symmetric pure-strategy equilibrium exists.

We first consider a Stackelberg model in which the firm commits to a hiring policy, under two distinct information structures. In the score-based model, the firm sequentially observes applicants' exact quality scores together with the full history of previously observed scores. We show that equilibrium arrival behavior exhibits a two-region structure: applicants whose scores lie in the firm's immediate acceptance set pool at time 0, while those outside pool at time 1. This partitioning enables the firm to infer each candidate's quality range from their arrival time, improving hiring outcomes. As the number of applicants grows large, numerical results indicate that the firm's optimal policy achieves a success probability of approximately 0.71, substantially exceeding the classical benchmark of 0.58. In the rank-based model, the firm sequentially observes each applicant's relative rank among those seen so far, together with the full history of past ranks. Under this information structure, equilibrium behavior features complete pooling: all applicants arrive at the same time, and after tie-breaking the resulting environment collapses to the classical secretary problem. The firm's success probability therefore coincides exactly with the classical benchmark of approximately 0.37.

We also analyze a simultaneous-move extension in which the firm lacks commitment power. In the rank-based model, equilibrium again reduces to the classical secretary problem. In the score-based model, the absence of commitment power is more severe: no symmetric BNE exists. The firm's commitment power is therefore essential for strategic arrival behavior to generate value.

The comparison between the two information models yields two central insights. First, information is essential. When the firm observes only ranks, applicants' strategic behavior creates no additional value: all applicants pool at a single arrival time, arrival order

remains uninformative, and the firm effectively faces the classical secretary problem. When the firm observes exact scores, strategic arrival becomes informative — applicants sort by quality through their timing choices, and arrival time conveys information that improves hiring outcomes. Second, commitment power is critical. Even when strategic timing is informative, the firm benefits only if it can credibly commit to a hiring policy in advance. Without such commitment, no symmetric equilibrium exists in the score-based model. Strategic behavior on the applicants' side alone is therefore insufficient; the firm must also possess sufficiently informative quality signals and credible commitment power for strategic arrival to improve hiring outcomes.

References

- Abhishek, Vibhanshu, Mustafa Dogan, Alexandre Jacquillat. 2021. Strategic timing and dynamic pricing for online resource allocation. *Management Science* **67**(8) 4880–4907.
- Alpern, Steve, Vic Baston. 2017. The secretary problem with a selection committee: Do conformist committees hire better secretaries? *Management Science* **63**(4) 1184–1197.
- Alpern, Steve, Shmuel Gal. 2009. Analysis and design of selection committees: a game theoretic secretary problem. *International Journal of Game Theory* **38**(3) 377–394.
- Alpern, Steve, Shmuel Gal, Eilon Solan. 2010. A sequential selection game with vetoes. *Games and Economic Behavior* **68**(1) 1–14.
- Bartoszyński, R, Z Govindarajulu. 1978. The secretary problem with interview cost. *Sankhyā: The Indian Journal of Statistics, Series B* 11–28.
- Bruss, F Thomas. 1984. A unified approach to a class of best choice problems with an unknown number of options. *The Annals of Probability* 882–889.
- Buchbinder, Niv, Kamal Jain, Mohit Singh. 2014. Secretary problems via linear programming. *Mathematics of Operations Research* **39**(1) 190–206.
- Caldentey, René, Gustavo Vulcano. 2007. Online auction and list price revenue management. *Management Science* **53**(5) 795–813.
- Chen, Yiwei, Vivek F Farias. 2018. Robust dynamic pricing with strategic customers. *Mathematics of Operations Research* **43**(4) 1119–1142.
- Chow, Yuan-Shih, Sigaiti Moriguti, Herbert Robbins, Stephen Mitchell Samuels. 1964. Optimal selection based on relative rank (the “secretary problem”). *Israel Journal of mathematics* **2**(2) 81–90.
- Chun, Young Hak. 1996. Selecting the best choice in the weighted secretary problem. *European journal of operational research* **92**(1) 135–147.
- Correa, José, Andrés Cristi, Laurent Feuilloley, Tim Oosterwijk, Alexandros Tsigonias-Dimitriadis. 2025. The secretary problem with independent sampling. *Management Science* **71**(4) 2778–2801.
- Cownden, Daniel, David Steinsaltz. 2014. Effects of competition in a secretary problem. *Operations Research* **62**(1) 104–113.

- DeGroot, Morris H. 2005. *Optimal statistical decisions*. John Wiley & Sons.
- Eriksson, Kimmo, Joel Sjöstrand, Pontus Strimling. 2007. Optimal expected rank in a two-sided secretary problem. *Operations Research* **55**(5) 921–933. doi:10.1287/opre.1070.0418.
- Ferguson, Thomas S. 1989. Who solved the secretary problem? *Statistical Science* **4**(3) 282–289. doi:10.1214/ss/1177012493.
- Freeman, Peter R. 1983. The secretary problem and its extensions: A review. *International Statistical Review* **51**(2) 189–206. doi:10.2307/1402748.
- Fujii, Kaito, Yuichi Yoshida. 2024. The secretary problem with predictions. *Mathematics of operations research* **49**(2) 1241–1262.
- Gianini, Jacqueline, Stephen M Samuels. 1976. The infinite secretary problem. *The Annals of Probability* 418–432.
- Gianini-Pettitt, Jacqueline. 1979. Optimal selection based on relative ranks with a random number of individuals. *Advances in Applied probability* **11**(4) 720–736.
- Gilbert, John P, Frederick Mosteller. 1966. Recognizing the maximum of a sequence. *Journal of the American Statistical Association* **61**(313) 35–73.
- Hill, Theodore P, Ulrich Krengel. 1991. Minimax-optimal stop rules and distributions in secretary problems. *The Annals of Probability* 342–353.
- Honnappa, Harsha, Rahul Jain. 2015. Strategic arrivals into queueing networks: The network concert queueing game. *Operations Research* **63**(1) 247–259.
- Hu, Ming, Yang Li, Jianfu Wang. 2018. Efficient ignorance: Information heterogeneity in a queue. *Management Science* **64**(6) 2650–2671.
- Immorlica, Nicole, Robert Kleinberg, Mohammad Mahdian. 2006. Secretary problems with competing employers. *International Workshop on Internet and Network Economics*. Springer, 389–400.
- Kakade, Sham M, Ilan Lobel, Hamid Nazerzadeh. 2013. Optimal dynamic mechanism design and the virtual-pivot mechanism. *Operations Research* **61**(4) 837–854.
- Karni, Edi, Aba Schwartz. 1977. Two theorems on optimal stopping with backward solicitation. *Journal of Applied Probability* **14**(4) 869–875.
- Kesselheim, Thomas, Alexandros Psomas, Shai Vardi. 2024. On hiring secretaries with stochastic departures. *Operations Research* **72**(5) 2076–2081.
- Kokkodis, Marios, Sam Ransbotham. 2023. Learning to successfully hire in online labor markets. *Management Science* **69**(3) 1597–1614.
- Lindley, Denis V. 1961. Dynamic programming and decision theory. *Journal of the Royal Statistical Society: Series C (Applied Statistics)* **10**(1) 39–51.
- Lorenzen, Thomas J. 1981. Optimal stopping with sampling cost: the secretary problem. *The Annals of Probability* **9**(1) 167–172.
- Moorhead, Cassie. 2024. The best time to apply for jobs, explained. <https://www.upwork.com/resources/best-time-to-apply-for-jobs>. Accessed: 2025-08-29.
- Mucci, Anthony G. 1973. On a class of secretary problems. *The Annals of Probability* 417–427.

- Papanastasiou, Yiangos, Nicos Savva. 2017. Dynamic pricing in the presence of social learning and strategic consumers. *Management Science* **63**(4) 919–939.
- Petrucelli, Joseph D. 1981. Best-choice problems involving uncertainty of selection and recall of observations. *Journal of Applied Probability* **18**(2) 415–425.
- Preater, John. 1994. On multiple choice secretary problems. *Mathematics of Operations Research* **19**(3) 597–602.
- Rocha, Amy L. 1993. The infinite secretary problem with recall. *The Annals of Probability* 898–916.
- Rose, John S. 1982. A problem of optimal choice and assignment. *Operations Research* **30**(1) 172–181.
- Rose, John S. 1984a. Optimal sequential selection based on relative ranks with renewable call options. *Journal of the American Statistical Association* **79**(386) 430–435.
- Rose, John S. 1984b. The secretary problem with a call option. *Operations research letters* **3**(5) 237–241.
- Salem, Jad, Swati Gupta. 2024. Secretary problems with biased evaluations using partial ordinal information. *Management Science* **70**(8) 5337–5366.
- Samuels, Stephen M. 1985. A best-choice problem with linear travel cost. *Journal of the American Statistical Association* **80**(390) 461–464.
- Smith, MH, JJ Deely. 1975. A secretary problem with finite memory. *Journal of the American Statistical Association* **70**(350) 357–361.
- Stewart, Theodor J. 1981. The secretary problem with an unknown number of options. *Operations Research* **29**(1) 130–145.
- Su, Xuanming. 2007. Intertemporal pricing with strategic customer behavior. *Management Science* **53**(5) 726–741.
- Tamaki, Mitsushi. 1979. A secretary problem with double choices. *Journal of the Operations Research Society of Japan* **22**(4) 257–265.
- Tamaki, Mitsushi. 1984. The secretary problem with optimal assignment. *Operations research* **32**(4) 847–858.
- Tamaki, Mitsushi. 1986. A full-information best-choice problem with finite memory. *Journal of applied probability* **23**(3) 718–735.
- Truong, Nguyen Le. 2013. Incentives and strategic choices in the secretary problem. Ph.D. thesis, University of California, Berkeley, Berkeley, CA. Doctoral dissertation.
- Upwork. 2024. Review job proposals. <https://support.upwork.com/hc/en-us/articles/18010402882195--Review-job-proposals>. Accessed: June 2026.
- Yang, Mark CK. 1974. Recognizing the maximum of a random sequence based on relative rank with backward solicitation. *Journal of Applied Probability* **11**(3) 504–512.
- Zhang, Junhui, Patrick Jaillet. 2023. Secretary problems with random number of candidates: How prior distributional information helps. *arXiv preprint arXiv:2310.07884* .
- Zorc, Saša, Ilia Tsetlin. 2020. Deadlines, offer timing, and the search for alternatives. *Operations Research* **68**(3) 927–948.

Appendix

A. Proofs

Proof of Proposition 1. Fix some $\pi \in \Pi_1^S \cup \Pi_1^R$. For notational brevity, define $W := W_0 \cap A^c$. Suppose, for a contradiction, that $\lambda(W) > 0$, which further implies that it is non-empty. Define $q_0 := \text{essinf } W$, so that by construction, we can always find some arbitrarily small $\epsilon > 0$ such that $\lambda((q_0, q_0 + \epsilon) \cap W) > 0$. Fix one such $\epsilon > 0$. We will focus on a generic applicant i with score $q_i(\epsilon) \in (q_0, q_0 + \epsilon) \cap W$ and note that by construction, this applicant arrives at $t^*(q_i(\epsilon), \pi) = 0$. For notational brevity, throughout this proof, write

$$p_i(t) := \mathbb{P}(\sigma_{(t^*(Q_1, \pi), \dots, t^*(Q_{i-1}, \pi), t, t^*(Q_{i+1}, \pi), \dots, t^*(Q_{N+1}, \pi))}(j_\pi^\dagger) = i | Q_i = q_i(\epsilon))$$

to represent applicant i 's acceptance probability, as a function of their chosen arrival time t . Finally, since $W \subseteq A^c$, we have $\pi_1(q_i(\epsilon)) = 0$, and therefore for i to be accepted, it is necessary for at least one other applicant to also choose $t = 0$.

Let $F_j, j \in \mathcal{N}$ represent the event that applicant i ends up in position j . Using the law of total probability, we can simplify $p_i(0)$ as follows

$$p_i(0) = \sum_{j=2}^{N+1} \mathbb{P}(\{j_\pi^\dagger = j\} \cap F_j | Q_i = q_i(\epsilon)). \quad (19)$$

Above, the sum omits $j = 1$ because $q_i(\epsilon) \in A^c$, which means that applicant i cannot be accepted in the first position. We now aim to construct an upper bound for $p_i(0)$ by bounding each of the terms in the summand on the RHS of (19).

Fix $j \in \{2, \dots, N\}$ and, for notational brevity, define $G := \{j_\pi^\dagger = j\} \cap F_j$. As an intermediate step, we aim to show that each of the following two conditional probabilities vanish:

$$\begin{aligned} \mathbb{P}(G \cap \{Q_{\sigma(1)} \leq q_0\} | Q_i = q_i(\epsilon)) &= 0, \quad \text{and} \\ \mathbb{P}(G \cap \{Q_{\sigma(1)} \geq q_0 + \epsilon\} | Q_i = q_i(\epsilon)) &= 0. \end{aligned} \quad (20)$$

For the first equation of (20), note that

$$\begin{aligned} &\mathbb{P}(G \cap \{Q_{\sigma(1)} \leq q_0\} | Q_i = q_i(\epsilon)) \\ &\leq \mathbb{P}(\{Q_{\sigma(1)} \leq q_0\} \cap \{Q_{\sigma(1)} \in W\} | Q_i = q_i(\epsilon)) + \mathbb{P}(G \cap \{Q_{\sigma(1)} \in W^c\} | Q_i = q_i(\epsilon)) \\ &\leq \mathbb{P}(\{Q_{\sigma(1)} \leq q_0\} \cap \{Q_{\sigma(1)} \in W\} | Q_i = q_i(\epsilon)) + \mathbb{P}(F_j \cap \{Q_{\sigma(1)} \in W_0^c\} | Q_i = q_i(\epsilon)) + \mathbb{P}(G \cap \{Q_{\sigma(1)} \in A\} | Q_i = q_i(\epsilon)) \\ &= 0. \end{aligned}$$

In the penultimate expression above, the first term vanishes because $q_0 = \text{essinf } W$ by definition. The second term vanishes because $j \geq 2$ and $t^*(q_i(\epsilon), \pi) = 0$ means that applicant i arrives at time 0 in position $j \geq 2$, so the first applicant must also arrive at time 0 and hence cannot have a score in W_0^c . Finally, the third term vanishes because on $\{Q_{\sigma(1)} \in A\}$ the first applicant is accepted in the first position, so applicant i cannot be hired in position $j \geq 2$.

For the second equation of (20), note that $q_0 + \epsilon > q_i(\epsilon)$, so we have

$$\mathbb{P}(G \cap \{Q_{\sigma(1)} \geq q_0 + \epsilon\} | Q_i = q_i(\epsilon)) \leq \mathbb{P}(G \cap \{Q_{\sigma(1)} \geq q_i(\epsilon)\} | Q_i = q_i(\epsilon)) = 0,$$

where the final equality is by construction of $\Pi_1^S \cup \Pi_1^R$ (see Lemma 1); since the first applicant has a score exceeding applicant i 's score, the latter cannot be hired in any position $j \geq 2$.

Therefore,

$$\begin{aligned}
\mathbb{P}(G | Q_i = q_i(\epsilon)) &= \mathbb{P}(G \cap \{Q_{\sigma(1)} \in (q_0, q_0 + \epsilon)\} | Q_i = q_i(\epsilon)) && [\text{By (20)}] \\
&\leq \mathbb{P}(Q_{\sigma(1)} \in (q_0, q_0 + \epsilon) | Q_i = q_i(\epsilon)) \\
&\leq \mathbb{P}\left(\bigcup_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in (q_0, q_0 + \epsilon)\} \middle| Q_i = q_i(\epsilon)\right) \\
&= \mathbb{P}\left(\bigcup_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in (q_0, q_0 + \epsilon)\}\right) && [(Q_1, \dots, Q_{N+1}) \text{ i.i.d.}] \\
&\leq N\epsilon. && [\text{Union bound}]
\end{aligned}$$

It remains to consider the case of $j = N + 1$. Since the focal applicant i arrives at time 0, the only way for this applicant to be accepted in the $(N + 1)$ th position is if (a) all other applicants also arrive at time 0, (b) all other applicants are rejected, and (c) applicant i is randomized into the $(N + 1)$ th position. We can represent this as follows:

$$\begin{aligned}
&\mathbb{P}(\{j_\pi^\dagger = N + 1\} \cap F_{N+1} | Q_i = q_i(\epsilon)) \\
&= \underbrace{\mathbb{P}\left(\left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{t^*(Q_m, \pi) = 0\}\right) \cap \left(\bigcap_{j=1}^N \{\pi_j(\mathbf{Q}_j) = 0\}\right) \middle| Q_i = q_i(\epsilon)\right)}_{:= \beta_\epsilon} \frac{1}{N + 1}. \tag{21}
\end{aligned}$$

Putting these together into (19), we obtain the upper bound:

$$p_i(0) \leq N(N - 1)\epsilon + \frac{\beta_\epsilon}{N + 1}. \tag{22}$$

Now suppose that applicant i with score $q_i(\epsilon)$ deviates and instead chooses $t_i = 1$. We now derive a lower bound for the probability that this applicant is accepted. Namely, the set of events under which applicant i is accepted contains the special case where (a) all other applicants choose to arrive at $t = 0$, (b) all other applicants are rejected, and (c) applicant i is accepted in the $(N + 1)$ th position. This gives rise to the following inequality

$$p_i(1) \geq \mathbb{P}\left(\left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{t^*(Q_m, \pi) = 0\}\right) \cap \left(\bigcap_{j=1}^N \{\pi_j(\mathbf{Q}_j) = 0\}\right) \middle| Q_i = q_i(\epsilon)\right) = \beta_\epsilon, \tag{23}$$

where the final equality above is by definition of β_ϵ . Comparing (23) to (21), we note that there is no $\frac{1}{N+1}$ term because applicant i , by choosing $t_i = 1$, is guaranteed to be in the final position.

Finally, note that by construction, since $W \subseteq W_0$,

$$\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in W\} \subseteq \bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{t^*(Q_m, \pi) = 0\}.$$

In addition, we also have

$$\{Q_{\sigma(1)} \in W\} \cap \bigcap_{j=2}^N \{Q_{\sigma(1)} > Q_{\sigma(j)}\} \subseteq \bigcap_{j=1}^N \{\pi_j(Q_j) = 0\},$$

because the event on the LHS ensures that (a) the first applicant is rejected, and (b) the second to N th applicants cannot be the global best. Putting both set relations together, and evaluating probabilities, we obtain from (23) that

$$\beta_\epsilon \geq \mathbb{P} \left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in W\} \cap \bigcap_{j=2}^N \{Q_{\sigma(1)} > Q_{\sigma(j)}\} \mid Q_i = q_i \right) = \frac{\lambda(W)^N}{N} > 0. \quad (24)$$

The equality above is a consequence of the i.i.d. assumption on the distribution of quality scores. It is important to note that the RHS of the inequality does not depend on ϵ .

Consequently, we can evaluate applicant i 's gain in utility from deviating to $t_i = 1$ from $t_i = 0$, as

$$\begin{aligned} p_i(1) - p_i(0) &\geq \frac{N}{N+1} \beta_\epsilon - N^2 \epsilon && [\text{By (22) and (23)}] \\ &\geq \frac{\lambda(W)^N}{N+1} - N^2 \epsilon && [\text{By (24)}] \\ &> 0 && [\text{for sufficiently small } \epsilon] \end{aligned}$$

Therefore, we have shown that if $\lambda(W) > 0$, there exists a set $(q_0, q_0 + \epsilon) \cap W$ with positive measure and an element of this set $q_i(\epsilon)$ such that applicant i with score $q_i(\epsilon)$ has strictly better utility by deviating from $t_i = 0$ to $t_i = 1$, contradicting the assumption of t^* being an equilibrium. Therefore, it is necessary that $\lambda(W) = 0$. $\square$

Proof of Proposition 2. Fix some $\pi \in \Pi_1^S \cup \Pi_1^R$, and focus on a generic applicant i with score $q_i \in A$. For notational brevity, throughout this proof we write

$$p_i(t) := \mathbb{P} \left(\sigma_{(t^*(Q_1, \pi), \dots, t^*(Q_{i-1}, \pi), t, t^*(Q_{i+1}, \pi), \dots, t^*(Q_{N+1}, \pi))} (j_\pi^\dagger) = i \mid Q_i = q_i \right)$$

for applicant i 's probability of acceptance as a function of their chosen arrival time t . In addition, as a preliminary observation, by the law of total probability and Proposition 1, we have

$$\lambda(W_0) = \lambda(W_0 \cap A). \quad (25)$$

We will suppose, for a contradiction, that $\lambda(W_0^c \cap A) > 0$.

Case 1: $\lambda(W_0) > 0$. Suppose that applicant i 's score $q_i \in W_0^c \cap A$, which is non-empty by assumption, and note that $q_i \in W_0^c$ means that applicant i 's arrival time, $t_i > 0$. A necessary (but generally not sufficient) condition for applicant i to be accepted is that no other applicant has a score in W_0 , because (25) implies that any such applicant would also almost surely be accepted at time 0 before applicant i . This implies that applicant i 's acceptance probability is

$$p_i(t_i) \leq (1 - \lambda(W_0))^N.$$

Now suppose that applicant i chooses to arrive at $t_i = 0$. Any other applicant who arrives at time $t = 0$ must have score in W_0 by construction. A consequence of (25) is that the first applicant (at time $t = 0$) will almost surely be accepted, because that applicant is almost surely in the immediate acceptance set. Therefore, we can characterize the acceptance probability of applicant i as

$$p_i(0) = \sum_{k=0}^N \binom{N}{k} \lambda(W_0)^k [1 - \lambda(W_0)]^{N-k} \frac{1}{k+1}.$$

In the summand above, k represents the number of *other* applicants who arrive at time $t = 0$; the probability of this occurring for each other applicant is $\lambda(W_0)$. The term $\frac{1}{k+1}$ appears because when there are k other applicants who choose time $t = 0$, the probability that applicant i is the first applicant is exactly $\frac{1}{k+1}$, since all $k+1$ applicants would be randomly sorted.

The above gives us

$$\begin{aligned} p_i(0) - p_i(t_i) &\geq \sum_{k=0}^N \frac{1}{k+1} \binom{N}{k} \lambda(W_0)^k [1 - \lambda(W_0)]^{N-k} - (1 - \lambda(W_0))^N \\ &= \sum_{k=1}^N \frac{1}{k+1} \binom{N}{k} \lambda(W_0)^k [1 - \lambda(W_0)]^{N-k} \\ &> 0. \end{aligned}$$

Thus we have constructed a set with positive measure (i.e., $W_0^c \cap A$), with an element q_i such that applicant i with score q_i strictly benefits from deviating to $t_i = 0$, establishing a contradiction.

Case 2: $\lambda(W_0) = 0$. Observe that for this case, we have $p_i(0) = 1$, because applicant i will almost surely be the first applicant by arriving at $t_i = 0$, and would be accepted because $q_i \in A$. Define

$$\hat{t} := \operatorname{ess\,sup}_{q \in A} t^*(q, \pi),$$

and note that since $\lambda(W_0^c \cap A) > 0$ by assumption, we must have $\hat{t} > 0$. Recalling the definition of W_t from (10), we define $\hat{\lambda} := \lambda(W_{\hat{t}} \cap A)$ and consider two sub-cases depending on whether $\hat{\lambda}$ is positive or not.

Sub-case 2A: $\hat{\lambda} > 0$. Suppose that applicant i 's score, $q_i \in \{q \in A : t^*(q, \pi) = \hat{t}\}$, i.e., their arrival time is $t_i = \hat{t}$. Define the event $G := \bigcap_{m \in N, m \neq i} \{Q_m \in (W_{\hat{t}} \cap A)\}$. On this event, (a) all applicants arrive simultaneously at time $\hat{t}$ and (b) applicant i is not accepted as long as the tie-breaking process ends up with one of applicant i 's opponents in the first position, which occurs with probability $\frac{N}{N+1}$. This argument implies $\frac{N}{N+1} \mathbb{P}(G) \leq 1 - p_i(t_i)$, which we can rewrite as

$$p_i(t_i) \leq 1 - \frac{N}{N+1} \mathbb{P}(G) = 1 - \frac{N}{N+1} \hat{\lambda}^N < 1 = p_i(0).$$

Therefore, we have constructed a set $\{q \in A : t^*(q, \pi) = \hat{t}\}$ with positive measure such that if applicant i has score q_i in this set and uses the purported best response t^* , they would have a strictly positive gain in utility from deviating to the arrival time of zero, establishing a contradiction.

Sub-case 2B: $\hat{\lambda} = 0$. For an arbitrary $\epsilon > 0$, define the lower level set

$$S_\epsilon := \{q \in [0, 1] : t^*(q, \pi) \leq \hat{t} - \epsilon\}.$$

Now, by construction of $\hat{t}$, for any ϵ , we always have $\lambda(A \cap S_\epsilon^c) > 0$. Suppose that applicant i has score $q_i(\epsilon) \in A \cap S_\epsilon^c$. Consider the event where all of i 's opponents have scores in $A \cap S_\epsilon$. In this case, one of them will be the first applicant, and because their score is in A , this applicant would be accepted in the first position. This implies $\mathbb{P}\left(\bigcap_{m \in \mathcal{N}, m \neq i} \{Q_m \in (A \cap S_\epsilon)\}\right) = \lambda(A \cap S_\epsilon)^N \leq 1 - p_i(t^*(q_i(\epsilon), \pi))$. Rearranging and taking limits on both sides, we get

$$\begin{aligned} \limsup_{\epsilon \downarrow 0} p_i(t^*(q_i(\epsilon), \pi)) &\leq 1 - \limsup_{\epsilon \downarrow 0} \lambda(A \cap S_\epsilon)^N \\ &= 1 - \lambda(A)^N && \text{[Monotone convergence theorem]} \\ &\leq 1 - \lambda(W_0^c \cap A)^N && [W_0^c \cap A \subseteq A] \\ &< 1. \end{aligned}$$

This implies that for a sufficiently small ϵ , when applicant i has score $q_i(\epsilon) \in A \cap S_\epsilon^c$, and chooses the arrival time $t_i = t^*(q_i(\epsilon), \pi)$, they have acceptance probability $p_i(t_i) < 1 = p_i(0)$. Since $\lambda(A \cap S_\epsilon^c) > 0$, we have established a contradiction. $\square$

Proof of Lemma 2. Suppose for a contradiction that for some policy $\pi \in \Pi_1^S \cup \Pi_1^R$ with immediate acceptance set A such that $\lambda(A^c) > 0$, there exists some best response $t^*(\cdot, \pi)$ that is strictly increasing almost everywhere on A^c , i.e., such that the inversion set $B^\dagger := \{(q, q') \in A^c \times A^c : q < q', t^*(q, \pi) \geq t^*(q', \pi)\}$ satisfies $\lambda^2(B^\dagger) = 0$. Recall that E_0 represents the event that all applicants have scores in A^c .

We focus on a generic applicant i . For notational brevity, define

$$p_i(t|q_i) := \mathbb{P}\left(\sigma_{(t^*(Q_1, \pi), \dots, t^*(Q_{i-1}, \pi), t, t^*(Q_{i+1}, \pi), \dots, t^*(Q_{N+1}, \pi))}(j_\pi^\dagger) = i \mid Q_i = q_i\right)$$

for applicant i 's acceptance probability as a function of their chosen arrival time t and score q_i , and similarly define

$$p_i(t \mid E_0, q_i) := \mathbb{P}\left(\sigma_{(t^*(Q_1, \pi), \dots, t^*(Q_{i-1}, \pi), t, t^*(Q_{i+1}, \pi), \dots, t^*(Q_{N+1}, \pi))}(j_\pi^\dagger) = i \mid Q_i = q_i, E_0\right).$$

Observe that for the relevant scores $q_i \in A^c$, applicant i cannot be accepted on E_0^c , because on E_0^c at least one other applicant has score in A , and by Corollary 1, such an applicant chooses to arrive at time 0 and is accepted ahead of i . Therefore,

$$p_i(t|q_i) = p_i(t \mid E_0, q_i) \mathbb{P}(E_0 \mid Q_i = q_i) = p_i(t \mid E_0, q_i) [1 - \lambda(A)]^N. \quad (26)$$

For each $j = 1, \dots, N$, define the set of ascending score histories by

$$\Delta_j := \{(q_1, \dots, q_j) \in (A^c)^j : q_1 \leq \dots \leq q_j\},$$

and the set of ascending histories rejected through position j by

$$B_j := \{(q_1, \dots, q_j) \in \Delta_j : \pi_1(q_1) = \pi_2(q_1, q_2) = \dots = \pi_j(q_1, \dots, q_j) = 0\}.$$

Our proof proceeds in two steps. First, as a preliminary, we establish that in equilibrium, the probability of accepting the $(N+1)$ th applicant must be zero. Second, we proceed to construct a profitable deviation for applicant i .

Step 1: Zero probability of accepting $(N+1)$ th applicant. Our first step is to construct an upper bound for applicant i 's equilibrium probability of acceptance conditional on E_0 for a particular choice of q_i . Define $\underline{t} := \text{ess inf}_{q \in A^c} t^*(q, \boldsymbol{\pi})$, and for any $\epsilon > 0$, define

$$S_\epsilon := \{q \in [0, 1] : t^*(q, \boldsymbol{\pi}) \geq \underline{t} + \epsilon\}.$$

By the definition of $\underline{t}$, we have $\lambda(A^c \cap S_\epsilon) > 0$. Suppose that applicant i 's score satisfies $q_i(\epsilon) \in A^c \cap S_\epsilon$. On the event $\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in A^c \cap S_\epsilon\}$, applicant i arrives first and is rejected. Hence

$$p_i(t^*(q_i(\epsilon), \boldsymbol{\pi}) \mid E_0, q_i(\epsilon)) \leq 1 - \left[\frac{\lambda(A^c \cap S_\epsilon)}{\lambda(A^c)} \right]^N.$$

Consider applicant i deviating to $t_i = 1$. Conditional on E_0 , the first N arrivals are exactly the other N applicants, and applicant i is the $(N+1)$ th applicant, with probability 1. In this case, the probability of i being accepted is the event that the first N applicants are rejected. Moreover, since t^* is strictly increasing almost everywhere on A^c , their arrival order almost surely coincides with their score order: conditional on E_0 , the probability that some pair $m \neq m'$ satisfies $Q_m < Q_{m'}$ and $t^*(Q_m, \boldsymbol{\pi}) \geq t^*(Q_{m'}, \boldsymbol{\pi})$ is at most $(N+1)N\lambda^2(B^\dagger)/\lambda(A^c)^2 = 0$. Therefore, the joint density of $(Q_{\sigma(1)}, \dots, Q_{\sigma(N)})$ conditional on E_0 is $N!/\lambda(A^c)^N$ on Δ_N . Hence,

$$\begin{aligned} p_i(t^*(q_i(\epsilon), \boldsymbol{\pi}) \mid E_0, q_i(\epsilon)) &\geq \mathbb{P}((Q_{\sigma(1)}, \dots, Q_{\sigma(N)}) \in B_N \mid E_0, Q_i = q_i(\epsilon)) && [t^* \text{ is a best response}] \\ &= \mathbb{P}((Q_{\sigma(1)}, \dots, Q_{\sigma(N)}) \in B_N \mid E_0) && [\text{Independence}] \\ &= \int_{B_N} \frac{N!}{\lambda(A^c)^N} dq_1 \cdots dq_N. \end{aligned}$$

Letting $\epsilon \downarrow 0$ on both sides yields

$$\int_{B_N} \frac{N!}{\lambda(A^c)^N} dq_1 \cdots dq_N \leq \limsup_{\epsilon \downarrow 0} p_i(t^*(q_i(\epsilon), \boldsymbol{\pi}) \mid E_0, q_i(\epsilon)) \leq 1 - \left[\frac{\lambda(A^c)}{\lambda(A^c)} \right]^N = 0, \quad (27)$$

where we have invoked the monotone convergence theorem to get $\lambda(A^c \cap S_\epsilon) \uparrow \lambda(A^c)$. Therefore, we have shown that

$$\lambda^N(B_N) = 0. \quad (28)$$

Now let us consider what happens when applicant i plays their equilibrium action $t^*(q_i(\epsilon), \boldsymbol{\pi})$. Conditional on E_0 , all $N+1$ applicants have scores in A^c , and their arrival order coincides with score order almost surely by assumption on t^* . In other words, the first N arrivals are almost surely the N smallest scores among the $N+1$ applicants. Therefore, the joint density of $(Q_{\sigma_{t^*}(1)}, \dots, Q_{\sigma_{t^*}(N)})$, conditional on E_0 , is $\frac{(N+1)!\lambda(A^c \cap [q_N, 1])}{\lambda(A^c)^{N+1}}$ on Δ_N . Thus

$$\mathbb{P}(j_{\boldsymbol{\pi}}^\dagger = N+1 \mid E_0) = \int_{B_N} \frac{(N+1)!\lambda(A^c \cap [q_N, 1])}{\lambda(A^c)^{N+1}} dq_1 \cdots dq_N = 0, \quad (29)$$

where the final equality follows from (28).

Step 2: Profitable deviation for applicant i . In this second step, we argue that there exists a subset of A^c (comprising “high” scores) that has positive Lebesgue measure and for which applicant i has a profitable

deviation. If $N = 1$, then $B_N = B_1$, so (28) gives $\lambda(A^c) = \lambda(B_1) = 0$, contradicting $\lambda(A^c) > 0$; hence we may assume $N \geq 2$ in what follows. Define

$$j^* := \min \{j \in \{2, \dots, N\} : \lambda^j(B_j) = 0\},$$

and note that j^* is well-defined since $\lambda(B_1) = \lambda(A^c) > 0$, while (28) gives $\lambda^N(B_N) = 0$. Let $\bar{q} := \text{esssup } A^c$. Further define

$$B_{j^*-1}^\delta := \{(q_1, \dots, q_{j^*-1}) \in B_{j^*-1} : q_{j^*-1} < \bar{q} - \delta\}.$$

Note that by the monotone convergence theorem, $\lim_{\delta \downarrow 0} \lambda^{j^*-1}(B_{j^*-1}^\delta) = \lambda^{j^*-1}(B_{j^*-1}) > 0$, which means that there must exist some $\delta > 0$ for which $\lambda^{j^*-1}(B_{j^*-1}^\delta) > 0$. Fix one such δ .

Suppose that applicant i has score $q_i \in A^c \cap (\bar{q} - \delta, \bar{q})$. Suppose that applicant i deviates to time $t^*(\bar{q} - \delta, \pi)$. Note that $\lambda^{j^*}(B_{j^*}) = 0$ by construction which implies that the position of the hired applicant is, almost surely, less than or equal to j^* . Since $A^c \cap (\bar{q} - \delta, \bar{q})$ has positive measure, for Lebesgue-a.e. $q_i \in A^c \cap (\bar{q} - \delta, \bar{q})$, the following event is a sufficient condition for applicant i to be accepted after deviating to $t^*(\bar{q} - \delta, \pi)$: (a) applicants 1 through $j^* - 1$ have (ordered) scores in $B_{j^*-1}^\delta$, and (b) all other $N - j^* + 1$ applicants have scores above $\bar{q} - \delta$. This is because in this event, both conditions (a) and (b) guarantee that i is the j^* th applicant, and condition (a) ensures that applicants 1 through $j^* - 1$ are rejected, leaving applicant i to be accepted as the j^* th applicant. Formally, we have

$$\begin{aligned} p_i(t^*(\bar{q} - \delta, \pi) \mid E_0, q_i) &\geq \mathbb{P} \left(\{(Q_{\sigma(1)}, \dots, Q_{\sigma(j^*-1)}) \in B_{j^*-1}^\delta\} \cap \bigcap_{k=j^*+1}^{N+1} \{Q_{\sigma(k)} \geq \bar{q} - \delta\} \mid E_0, Q_i = q_i \right) \\ &= \binom{N}{j^*-1} (j^*-1)! \frac{\lambda^{j^*-1}(B_{j^*-1}^\delta) \lambda(A^c \cap (\bar{q} - \delta, \bar{q}))^{N-j^*+1}}{\lambda(A^c)^N} \\ &> 0, \end{aligned} \quad (30)$$

On the other hand, if applicant i plays their equilibrium action, (29) implies that a necessary condition for acceptance is that applicant i is not in position $N + 1$. This yields the upper bound

$$p_i(t^*(q_i, \pi) \mid E_0, q_i) \leq 1 - \left[\frac{\lambda(A^c \cap (0, q_i))}{\lambda(A^c)} \right]^N, \quad (31)$$

where the latter expression is a consequence of our assumption on t^* being strictly increasing a.e. on A^c .

We conclude the proof by observing that by taking $q_i \uparrow \bar{q}$ in (31), we can make $p_i(t^*(q_i, \pi) \mid E_0)$ arbitrarily close to zero, whereas (30) implies that $p_i(t^*(\bar{q} - \delta, \pi) \mid E_0)$ is bounded away from zero for all q_i . By (26), this implies that $t^*(\bar{q} - \delta, \pi)$ is a profitable deviation for all q_i sufficiently close to $\bar{q}$, and we can easily see that such q_i comprise a set of nonzero measure. This contradicts the assumption that $t^*(\cdot, \pi)$ is a best response. Therefore, in equilibrium, t^* cannot be strictly increasing a.e. on A^c . $\square$

Proof of Lemma 3. Fix $\pi \in \Pi_1^S \cup \Pi_1^R$ and write $t^*(\cdot) := t^*(\cdot, \pi)$ for brevity. By Lemma 2, t^* is not strictly increasing a.e. on A^c , and therefore the set $B := \{(q, q') \in A^c \times A^c : q < q', t^*(q) \geq t^*(q')\}$ satisfies $\lambda^2(B) > 0$ by definition. For each $m \in A^c$, define the slice

$$B_m := \{q \in A^c \cap [0, m) : t^*(q) \geq t^*(m)\}. \quad (32)$$

By Fubini's theorem,

$$\int_{A^c} \lambda(B_m) dm = \lambda^2(B) > 0. \quad (33)$$

Let $M := \max\{Q_1, \dots, Q_N\}$ and let i^* be such that $Q_{i^*} = M$. Define the event $E := \bigcap_{n=1}^N \{Q_n \in A^c\}$, and denote by $F_M(\cdot | E)$ the conditional distribution of M given E . Since $Q_1, \dots, Q_N$ are i.i.d. $U[0, 1]$, conditional on E they are i.i.d. uniform on A^c . Hence, for any $m \in [0, 1]$,

$$F_M(m | E) = \mathbb{P}(M \leq m | E) = \left(\frac{\lambda(A^c \cap [0, m])}{\lambda(A^c)} \right)^N.$$

Therefore M has conditional density

$$f_M(m | E) = \frac{N}{\lambda(A^c)} \left(\frac{\lambda(A^c \cap [0, m])}{\lambda(A^c)} \right)^{N-1} \mathbf{1}_{A^c}(m).$$

In particular, $f_M(m | E) > 0$ for a.e. $m \in A^c$. Equivalently, every measurable subset of A^c with positive Lebesgue measure receives strictly positive conditional probability under $F_M(\cdot | E)$.

Next observe that

$$\bigcap_{i \neq i^*} \{(Q_i, M) \in B\} \subseteq \bigcap_{i=1}^N \{t^*(Q_i) \geq t^*(M)\} = \left\{ t^*(M) = \min_{1 \leq i \leq N} t^*(Q_i) \right\}. \quad (34)$$

Therefore,

$$\begin{aligned} \mathbb{P}\left(t(M) = \min_{1 \leq i \leq N} t(Q_i) \middle| E\right) &\geq \int_{A^c} \mathbb{P}\left(\bigcap_{i \neq i^*} \{(Q_i, m) \in B\} \middle| M = m, E\right) dF_M(m | E) \quad [\text{by (34)}] \\ &= \int_{A^c} [\mathbb{P}((Q_1, m) \in B | Q_1 < m, E)]^{N-1} dF_M(m | E) \quad [(Q_1, \dots, Q_N) \text{ i.i.d.}] \\ &= \int_{A^c} \left[\frac{\lambda(B_m)}{\lambda(A^c \cap [0, m])} \right]^{N-1} dF_M(m | E), \quad [\text{by (32)}] \\ &\geq \int_{A^c} [\lambda(B_m)]^{N-1} dF_M(m | E) \quad [\lambda(A^c \cap [0, m]) \leq 1] \\ &\geq \left(\int_{A^c} \lambda(B_m) dF_M(m | E) \right)^{N-1} \quad [\text{Jensen's inequality}] \\ &> 0, \end{aligned}$$

where the final strict inequality follows from (33) and the fact that every measurable subset of A^c with positive Lebesgue measure receives strictly positive conditional probability under $F_M(\cdot | E)$. $\square$

Proof of Theorem 1. Fix some $\pi \in \Pi_1^S \cup \Pi_1^R$ with corresponding immediate acceptance set A . Let t^* represent a symmetric equilibrium response by the applicants. In this proof, we will first establish that the structure (13) is necessary for an equilibrium response, and we will conclude the proof by constructively demonstrating that it is also sufficient. Because the structure (13) uniquely determines applicants' responses (up to sets of zero measure), uniqueness follows as a consequence of the second constructive step.

We note a few preliminaries before we begin the proof. First, note that $t^*(q, \pi) = 0$ for almost every $q \in A$ is a consequence of Corollary 1. Second, note that if $\lambda(A^c) = 0$, the result holds automatically. Therefore, we henceforth assume the existence of a set $B \subseteq A^c$ with $\lambda(B) > 0$ such that $0 < t^*(q, \pi) < 1$ for all $q \in B$, and proceed to prove the necessity part of the theorem by contradiction. Our proof will focus on a generic

applicant i whose quality score is in B , and demonstrate that they can strictly increase their utility by deviating to choosing the arrival time of 1.

We begin by establishing some notation. Recall that E_0 represents the event that no applicants have scores in A , and note that (11), together with the assumption that $\lambda(A^c) \geq \lambda(B) > 0$ implies that $\mathbb{P}(E_0) > 0$. For notational brevity, we will write

$$p_i(t) := \mathbb{P} \left(\sigma_{(t^*(Q_1, \pi), \dots, t^*(Q_{i-1}, \pi), t, t^*(Q_{i+1}, \pi), \dots, t^*(Q_{N+1}, \pi))} (j_\pi^\dagger) = i \mid Q_i = q_i \right)$$

to represent the probability of applicant i being accepted as a function of their chosen time t , and similarly write

$$p_i(t|E_0) := \mathbb{P} \left(\sigma_{(t^*(Q_1, \pi), \dots, t^*(Q_{i-1}, \pi), t, t^*(Q_{i+1}, \pi), \dots, t^*(Q_{N+1}, \pi))} (j_\pi^\dagger) = i \mid E_0, Q_i = q_i \right)$$

to represent the analogous probability with a further conditioning on E_0 . We observe that $p_i(t) = p_i(t|E_0)\mathbb{P}(E_0 \mid Q_i = q_i)$, where $\mathbb{P}(E_0 \mid Q_i = q_i) = [1 - \lambda(A)]^N > 0$ for $q_i \in A^c$, because on the event E_0^c , at least one of i 's opponents has a score in A , almost surely arrives at time 0 by Corollary 1, and is accepted by construction, which means that i has zero probability of acceptance on E_0^c . Therefore, to establish a contradiction that t^* is a best response, it suffices to exhibit some subset $\mathcal{Q} \subseteq [0, 1]$ with positive measure, such that for $q_i \in \mathcal{Q}$, we have

$$p_i(1|E_0) - p_i(t^*(q_i, \pi)|E_0) > 0. \quad (35)$$

To do so, we define $\underline{t} := \text{ess inf}_{q \in A^c} t^*(q, \pi)$. We may assume that $\underline{t} < 1$, otherwise, the theorem holds vacuously. We also define the set $\underline{B} := W_{\underline{t}} \cap A^c$. Our proof proceeds by considering two cases.

Case 1: $\lambda(\underline{B}) = 0$: For $\epsilon > 0$, define the upper level set

$$S_\epsilon := \{q \in [0, 1] : t^*(q, \pi) \geq \underline{t} + \epsilon\},$$

By construction of $\underline{t}$, we have $\lambda(A^c \cap S_\epsilon) > 0$ for all $\epsilon > 0$. Consider applicant i with score $q_i(\epsilon) \in A^c \cap S_\epsilon$. The event $\bigcap_{m \in \mathcal{N}, m \neq i} \{Q_m \in A^c \cap S_\epsilon\}$ is a sufficient condition for applicant i to be rejected, because on this event, applicant i is in the first position, and is rejected because $q_i(\epsilon) \in A^c$. This implies

$$p_i(t^*(q_i(\epsilon), \pi)|E_0) \leq 1 - \mathbb{P} \left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in S_\epsilon\} \mid E_0 \right) = 1 - \left[\frac{\lambda(S_\epsilon)}{\lambda(A^c)} \right]^N.$$

Letting $\epsilon \downarrow 0$ on both sides of the inequality gives us

$$\limsup_{\epsilon \downarrow 0} p_i(t^*(q_i(\epsilon), \pi) \mid E_0) \leq 1 - \limsup_{\epsilon \downarrow 0} \left[\frac{\lambda(A^c \cap S_\epsilon)}{\lambda(A^c)} \right]^N = 1 - \left[\frac{\lambda(A^c)}{\lambda(A^c)} \right]^N = 0, \quad (36)$$

where the second equality above is a consequence of the monotone convergence theorem.

Next, we compute applicant i 's probability of being accepted if they deviate to time $t_i = 1$. Define event D as follows

$$D := \{t^*(\max\{Q_1, \dots, Q_{i-1}, Q_{i+1}, \dots, Q_{N+1}\}, \pi) = \min\{t^*(Q_1, \pi), \dots, t^*(Q_{i-1}, \pi), t^*(Q_{i+1}, \pi), \dots, t^*(Q_{N+1}, \pi)\}\},$$

which represents the case where the opponent with the highest score arrives earliest among them. Lemma 3 shows that we have $\mathbb{P}(D|E_0) > 0$.

Consider now the event $D \cap \{\sigma(N+1) = i\} \cap E_0$, which we claim is sufficient for i to be accepted. The first applicant is rejected because their score is in A^c , and the second through the N th applicants are rejected because, among the first N applicants, the first applicant already had the highest score (by construction of Π_1^S and Π_1^R , see Lemma 1). Applicant i , being the final applicant, is accepted since $\pi_{N+1} = 1$, independent of their realized score $q_i(\epsilon)$. Therefore, conditioning on E_0 , we have shown that

$$p_i(1|E_0) \geq \mathbb{P}(D \cap \{\sigma(N+1) = i\} | E_0).$$

We now argue that

$$\mathbb{P}(\{\sigma(N+1) = i\} | D, E_0) > 0.$$

This is because when applicant i arrives at the latest possible time of $t_i = 1$, conditional on D and E_0 , they will always have some positive probability of being in the last position as (a) ties are broken uniformly at random, and (b) there are only finitely many applicants. Therefore, putting these together, we have shown that

$$p_i(1|E_0) \geq \mathbb{P}(D \cap \{\sigma(N+1) = i\} | E_0) = \mathbb{P}(\{\sigma(N+1) = i\} | D, E_0) \mathbb{P}(D|E_0) > 0.$$

Together with (36), this implies that (35) holds for sufficiently small ϵ .

Case 2: $\lambda(\underline{B}) > 0$: In this case, $\underline{B}$ is non-empty and we define $q_0 := \text{ess inf } \underline{B}$. By construction, for any $\epsilon > 0$, we have $\lambda([q_0, q_0 + \epsilon) \cap \underline{B}) > 0$. Let applicant i have score $q_i(\epsilon) \in [q_0, q_0 + \epsilon) \cap \underline{B}$, so that $t^*(q_i(\epsilon), \boldsymbol{\pi}) = \underline{t}$.

Let $F_j, j \in \mathcal{N}$ represent the event that applicant i ends up in position j . Using the law of total probability, we can simplify the conditional probability that applicant i is accepted, as follows

$$p_i(\underline{t}|E_0) = \sum_{j=2}^{N+1} \mathbb{P}\left(\{j_{\boldsymbol{\pi}}^{\dagger} = j\} \cap F_j \mid E_0, Q_i = q_i(\epsilon)\right). \quad (37)$$

Above, the sum omits $j = 1$ because $q_i(\epsilon) \in \underline{B} \subseteq A^c$, which means that applicant i cannot be accepted in the first position. We now aim to construct an upper bound for $p_i(\underline{t}|E_0)$ by bounding each of the terms in the summand on the RHS of (37).

Fix $j \in \{2, \dots, N\}$ and, for notational brevity, define $G := \{j_{\boldsymbol{\pi}}^{\dagger} = j\} \cap F_j$. As an intermediate step, we aim to show that each of the following two conditional probabilities vanish:

$$\begin{aligned} \mathbb{P}(G \cap \{Q_{\sigma(1)} \leq q_0\} \mid E_0, Q_i = q_i(\epsilon)) &= 0, \quad \text{and} \\ \mathbb{P}(G \cap \{Q_{\sigma(1)} \geq q_0 + \epsilon\} \mid E_0, Q_i = q_i(\epsilon)) &= 0. \end{aligned} \quad (38)$$

For the first equation of (38), note that

$$\begin{aligned} &\mathbb{P}(G \cap \{Q_{\sigma(1)} \leq q_0\} \mid E_0, Q_i = q_i(\epsilon)) \\ &\leq \mathbb{P}(\{Q_{\sigma(1)} \leq q_0\} \cap \{Q_{\sigma(1)} \in \underline{B}\} \mid E_0, Q_i = q_i(\epsilon)) + \mathbb{P}(G \cap \{Q_{\sigma(1)} \in \underline{B}^c\} \mid E_0, Q_i = q_i(\epsilon)) \\ &\leq \mathbb{P}(\{Q_{\sigma(1)} \leq q_0\} \cap \{Q_{\sigma(1)} \in \underline{B}\} \mid E_0, Q_i = q_i(\epsilon)) + \mathbb{P}(F_j \cap \{Q_{\sigma(1)} \in W_{\underline{t}}^c\} \mid E_0, Q_i = q_i(\epsilon)) \\ &\quad + \mathbb{P}(G \cap \{Q_{\sigma(1)} \in A\} \mid E_0, Q_i = q_i(\epsilon)) \end{aligned}$$

= 0.

In the penultimate expression, the first term vanishes because $q_0 = \text{essinf } \underline{B}$. The second term vanishes because $j \geq 2$ and $t^*(q_i(\epsilon), \pi) = \underline{t}$ imply that, on $E_0 \cap \{Q_i = q_i(\epsilon)\}$, if applicant i is in position j , then the first applicant cannot have a score in $W_{\underline{t}}^c$. The third term vanishes because, on $\{Q_{\sigma(1)} \in A\}$, the first applicant is accepted in the first position, so applicant i cannot be hired in any position $j \geq 2$.

For the second equation of (38), since $q_0 + \epsilon > q_i(\epsilon)$, we have

$$\mathbb{P}(G \cap \{Q_{\sigma(1)} \geq q_0 + \epsilon\} \mid E_0, Q_i = q_i(\epsilon)) \leq \mathbb{P}(G \cap \{Q_{\sigma(1)} > q_i(\epsilon)\} \mid E_0, Q_i = q_i(\epsilon)) = 0.$$

The final equality is by the construction of $\pi \in \Pi_1^S \cup \Pi_1^R$: conditional on E_0 and $Q_i = q_i(\epsilon)$, if the first applicant has a score strictly exceeding applicant i 's score, then applicant i cannot be hired in any position $j \geq 2$.

Therefore,

$$\begin{aligned} \mathbb{P}(G \mid E_0, Q_i = q_i(\epsilon)) &= \mathbb{P}(G \cap \{Q_{\sigma(1)} \in (q_0, q_0 + \epsilon)\} \mid E_0, Q_i = q_i(\epsilon)) && [\text{By (38)}] \\ &\leq \mathbb{P}(Q_{\sigma(1)} \in (q_0, q_0 + \epsilon) \mid E_0, Q_i = q_i(\epsilon)) \\ &\leq \mathbb{P}\left(\bigcup_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in (q_0, q_0 + \epsilon)\} \mid E_0, Q_i = q_i(\epsilon)\right) \\ &= \mathbb{P}\left(\bigcup_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in (q_0, q_0 + \epsilon)\} \mid E_0\right) && [(Q_1, \dots, Q_{N+1}) \text{ i.i.d.}] \\ &\leq \frac{N\epsilon}{\mathbb{P}(E_0)}. && [\text{Union bound}] \end{aligned}$$

For $j = N + 1$, it is easy to see that

$$\mathbb{P}\left(F_{N+1} \mid E_0, Q_i = q_i(\epsilon)\right) = \frac{1}{N+1} \mathbb{P}\left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in \underline{B}\} \mid E_0\right) = \frac{1}{N+1} \left(\frac{\lambda(\underline{B})}{1 - \lambda(A)}\right)^N$$

because on the set E_0 , applicant i will be in position $N + 1$ if, and only if, (a) all other applicants have scores in $\underline{B}$ so that they choose time $\underline{t}$ as well, and (b) the uniformly at random tie-breaking across the $N + 1$ applicants assigns applicant i to position $N + 1$. Putting these together, we obtain the upper bound:

$$p_i(\underline{t} \mid E_0) \leq \frac{N(N-1)}{\mathbb{P}(E_0)} \epsilon + \frac{1}{N+1} \left(\frac{\lambda(\underline{B})}{1 - \lambda(A)}\right)^N \quad (39)$$

Now suppose applicant i deviates to $t_i = 1$. On the event E_0 , a sufficient condition for this applicant to be accepted is if (a) all other applicants arrive at time $\underline{t}$, and (b) the first applicant has the highest quality among all the other applicants. The first applicant is rejected because their score is in A^c , and the second through the N th applicants are rejected because, among the first N applicants, the first applicant already had the highest score by Lemma 1. We can therefore derive a lower bound for the probability that this applicant is accepted, as follows

$$\begin{aligned}
p_i(1|E_0) &\geq \mathbb{P} \left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_m \in \underline{B}\} \cap \left\{ Q_{\sigma(1)} = \max_{\substack{m \in \mathcal{N} \\ m \neq i}} Q_m \right\} \middle| E_0, Q_i = q_i(\epsilon) \right) \\
&= \frac{1}{N} \mathbb{P} \left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_{\sigma(m)} \in \underline{B}\} \middle| E_0, Q_i = q_i(\epsilon) \right) && \text{[Uniform at random tie-breaking]} \\
&= \frac{1}{N} \mathbb{P} \left(\bigcap_{\substack{m \in \mathcal{N} \\ m \neq i}} \{Q_{\sigma(m)} \in \underline{B}\} \middle| E_0 \right) && [(Q_1, \dots, Q_{N+1}) \text{ i.i.d.}] \\
&= \frac{1}{N} \left(\frac{\lambda(\underline{B})}{1 - \lambda(A)} \right)^N.
\end{aligned} \tag{40}$$

From (40) and (39), applicant i 's gain in utility from deviating can be bounded by

$$p_i(1|E_0) - p_i(t_i|E_0) \geq \frac{1}{N} \left(\frac{\lambda(\underline{B})}{1 - \lambda(A)} \right)^N - \frac{1}{N+1} \left(\frac{\lambda(\underline{B})}{1 - \lambda(A)} \right)^N - \frac{N(N-1)}{\mathbb{P}(E_0)} \epsilon$$

and therefore (35) again holds for sufficiently small ϵ .

Thus far, we have shown that (13) is necessary for a symmetric best response among applicants. We now show that it is also sufficient. To be precise, consider the candidate strategy $t^*(q, \pi) = \mathbb{1}\{q \in A^c\}$. Fix applicant i with score q_i , and suppose that all N opponents use this candidate strategy. Let k denote the number of applicant i 's opponents whose scores lie in A .

First, consider any $q_i \in A$. If applicant i chooses $t_i = 0$, then

$$p_i(0) = \sum_{k=0}^N \binom{N}{k} \lambda(A)^k [1 - \lambda(A)]^{N-k} \frac{1}{k+1},$$

where the factor of $\frac{1}{k+1}$ is because i is randomly ordered among the $k+1$ applicants with scores in A . If instead $t_i > 0$, then a necessary condition for acceptance is that no opponent has a score in A , so

$$p_i(t_i) \leq [1 - \lambda(A)]^N \leq p_i(0),$$

and therefore $t^*(q_i, \pi) = 0$ is a best response for all $q_i \in A$.

Next, consider any $q_i \in A^c$. If applicant i chooses any $t_i < 1$, then

$$p_i(t_i) = p_i(t_i | E_0) \mathbb{P}(E_0 | Q_i = q_i) = 0,$$

where the first equality is because if there exists at least one opponent with score in A who arrives before applicant i , that opponent is accepted before i is reached. In turn, the second equality is because on E_0 , applicant i arrives before all such opponents but is rejected because $q_i \in A^c$, leading to $p_i(t_i | E_0) = 0$. Therefore, $p_i(1) \geq p_i(t_i) = 0$, and so $t^*(q_i, \pi) = 1$ is a best response for every $q_i \in A^c$, concluding the proof that $t^*(\cdot, \pi)$ is a symmetric equilibrium. $\square$

Proof of Theorem 2. Consider a generic $\pi \in \Pi_1^S$ with corresponding immediate acceptance set A . Let t^* represent a symmetric equilibrium response by the applicants, whose structure is specified by Theorem 1.

We recall the decomposition of the firm's objective by (12) from which we make three key observations. First, from (11), for each $k \geq 0$, $\mathbb{P}(E_k)$ only depends on π through the value of $\lambda(A)$. Second, from (12),

we observe that $\mathbb{P}(R_1 = 1|E_k)$ only depends on π through A but is otherwise independent of $\pi_2, \dots, \pi_N$. Indeed, we can actually write $\mathbb{P}(R_1 = 1|E_k)$ explicitly as

$$\mathbb{P}(R_1 = 1|E_k) = \int_A \frac{[\lambda(\{[0, q] \cap A^c\})]^{N+1-k} [\lambda(\{[0, q] \cap A\})]^{k-1}}{\lambda(A)^k [1 - \lambda(A)]^{N+1-k}} dq.$$

Third, on the event E_0 , all applicants choose $t^* = 1$ and are ordered uniformly at random, with scores i.i.d. uniform on A^c ; the first applicant is rejected with probability one since its score lies in A^c . Because the firm knows A , it may apply the quantile transform $q \mapsto \lambda(A^c \cap [0, q])/\lambda(A^c)$, under which the conditional scores are i.i.d. standard uniform and the identity of the best applicant is preserved. Hence the *optimal value* $\sup_{(\pi_2, \dots, \pi_N)} \mathbb{P}(R_{j_\pi^\dagger} = 1 | E_0)$ is a constant independent of the structure of A .

Our three observations above imply that we can reformulate the firm's optimization problem as

$$\sup_{\pi \in \Pi_1^S} \mathbb{P}(R_{j_\pi^\dagger} = 1) = \sup_{L \in [0, 1]} \Phi(L),$$

where we define

$$\Phi(L) := \mathbb{P}(E_0) \left(\sup_{(\pi_2, \dots, \pi_N)} \mathbb{P}(R_{j_\pi^\dagger} = 1 | E_0) \right) + \left(\sup_{A: \lambda(A) = 1-L} \sum_{k=1}^{N+1} \mathbb{P}(E_k) \mathbb{P}(R_1 = 1 | E_k) \right). \quad (41)$$

We now analyze each of the optimization problems of (41), beginning with the latter. Note that for $k \geq 1$, we have

$$\mathbb{P}(R_1 = 1 | E_k) \leq \frac{1}{k},$$

because of uniform at random tie-breaking of the k applicants at time $t = 0$. Indeed, we will show that for any $k \geq 1$, this upper bound is attained for every k by A having an upper interval structure, that is, $A = [L, 1]$. To see this, consider

$$\begin{aligned} \int_A \frac{[\lambda(\{[0, q] \cap A^c\})]^{N+1-k} [\lambda(\{[0, q] \cap A\})]^{k-1}}{\lambda(A)^k [1 - \lambda(A)]^{N+1-k}} dq &= \int_L^1 \frac{L^{N+1-k} (q-L)^{k-1}}{(1-L)^k L^{N+1-k}} dq \\ &= \frac{1}{(1-L)^k} \int_L^1 (q-L)^{k-1} dq \\ &= \frac{1}{k}. \end{aligned}$$

Note that the above holds for all $k \geq 1$ and any $L \in [0, 1]$. Therefore, we have shown that

$$\Phi(L) = \mathbb{P}(E_0) \left(\sup_{(\pi_2, \dots, \pi_N)} \mathbb{P}(R_{j_\pi^\dagger} = 1 | E_0) \right) + \sum_{k=1}^{N+1} \frac{1}{k} \mathbb{P}(E_k), \quad (42)$$

and where the second term is maximized by the choice of $A = [L, 1]$.

Now consider the remaining optimization problem in (42). On E_0 , the firm faces the full-information game of Gilbert and Mosteller (1966) with $N+1$ applicants whose scores are i.i.d. uniform on $[0, L]$, subject to the constraint that the first arrival is rejected regardless of its score. Since the constraint fixes only the first-position action, the backward induction determining the optimal continuation is unchanged from positions $2, \dots, N+1$: at each position j , it is optimal to accept if and only if the current score exceeds all previous scores and the indifference threshold $L\theta_j$, where θ_j is the decision number of Gilbert and Mosteller (1966, eq. (3b-1)) for position j of $N+1$, scaled by L via the scale invariance of the uniform distribution.

This is precisely the policy (14). Therefore, for every fixed $L \in [0, 1]$, a policy of the stated form is optimal, and the firm's problem reduces to the one-dimensional optimization over L . $\square$

Proof of Proposition 3. Given the firm's policy, we can specialize (12) to

$$\mathbb{P}(R_{j_\pi^\dagger} = 1) = \mathbb{P}(R_{j_\pi^\dagger} = 1 | E_0) \mathbb{P}(E_0) + \sum_{k=1}^{N+1} \frac{1}{k} \mathbb{P}(E_k),$$

where we have applied $\mathbb{P}(R_1 = 1 | E_k) = \frac{1}{k}$ for $k \geq 1$ because on E_k , the k applicants whose scores exceed L are randomly ordered after tie-breaking, and the firm hires the one who is in the first position. Hence, conditional on E_k , the firm selects the best among them with probability $1/k$.

If $k = 0$, all $N + 1$ applicants have scores i.i.d. uniform on $[0, L]$ and arrive in random order. As established in the proof of Theorem 2, the firm's problem is then the normalized full-information game of Gilbert and Mosteller (1966) with $N + 1$ applicants under the monotone decision-number strategy $(1, \theta_2, \dots, \theta_{N+1})$, where the first decision number of 1 encodes the forced rejection of the first arrival. Summing the win probabilities of equation (3c-1) in Gilbert and Mosteller (1966) over positions $2, \dots, N + 1$ yields the expression of Γ_N in the proposition statement, which is independent of L by scale invariance.

The expression for $\Phi(L)$ is a consequence of using the expression for $\mathbb{P}(E_k)$ from (11) for the case that $A = [L, 1]$. We now proceed to show that Φ has a unique maximizer that resides in the interior of $(0, 1)$.

Let $x = (1 - L)/L$, $x \in [0, \infty)$. Then

$$\Phi(L) = L^{N+1} \left[\Gamma_N + \sum_{k=1}^{N+1} \frac{1}{k} \binom{N+1}{k} x^k \right].$$

Differentiating gives

$$\begin{aligned} \Phi'(L) &= (N+1)L^N \left[\Gamma_N + \sum_{k=1}^{N+1} \frac{1}{k} \binom{N+1}{k} x^k \right] + L^{N+1} \left[\sum_{k=1}^{N+1} \binom{N+1}{k} x^{k-1} \left(-\frac{1}{L^2} \right) \right] \\ &= (N+1)L^N \left[\Gamma_N - \underbrace{\left(\frac{1+x}{N+1} \sum_{k=1}^{N+1} \binom{N+1}{k} x^{k-1} - \sum_{k=1}^{N+1} \frac{1}{k} \binom{N+1}{k} x^k \right)}_{=: g(x)} \right]. \end{aligned}$$

Then

$$\begin{aligned} g'(x) &= \frac{1}{N+1} \sum_{k=2}^{N+1} \binom{N+1}{k} (k-1) x^{k-2} + \frac{1}{N+1} \sum_{k=1}^{N+1} \binom{N+1}{k} k x^{k-1} - \sum_{k=1}^{N+1} \binom{N+1}{k} x^{k-1} \\ &= \frac{1}{N+1} \sum_{k=1}^N \binom{N+1}{k+1} k x^{k-1} + \frac{1}{N+1} \sum_{k=1}^N \binom{N+1}{k} k x^{k-1} - \sum_{k=1}^N \binom{N+1}{k} x^{k-1} \quad [\text{reindex, eliminate}] \\ &= \frac{1}{N+1} \sum_{k=1}^N \left[k \binom{N+1}{k+1} + k \binom{N+1}{k} - (N+1) \binom{N+1}{k} \right] x^{k-1} \\ &= \frac{1}{N+1} \sum_{k=1}^N [k - (k+1)] \binom{N+1}{k+1} x^{k-1} \\ &< 0. \end{aligned}$$

Hence g is strictly decreasing on $[0, \infty)$. Since $x = (1 - L)/L$ is strictly decreasing in L , $h(L) := g((1 - L)/L)$ is strictly increasing in L . Moreover, we have $\lim_{L \rightarrow 0} h(L) = -\infty < \Gamma_N$ and $h(1) = 1 > \Gamma_N$. Continuity of

h , together with the properties above imply that there exists a unique $L^* \in (0, 1)$ such that $h(L^*) = \Gamma_N$. It follows that $\Phi'(L) > 0$ for $L < L^*$ and $\Phi'(L) < 0$ for $L > L^*$. It remains to rule out $L = 0$. Evaluating at $L = 0$, we have $\Phi(0) = 1/(N+1)$. As $L \downarrow 0$,

$$\Phi(L) = \frac{1}{N+1} + \frac{1}{N}L + O(L^2) = \Phi(0) + \frac{1}{N}L + O(L^2),$$

where the expansion follows by retaining the terms with $k = N+1$ and $k = N$ in the sum; the terms with $k \leq N-1$ and the term $L^{N+1}\Gamma_N$ are $O(L^2)$. Thus $\Phi(L) > \Phi(0)$ for all sufficiently small $L > 0$, so $L = 0$ cannot be a maximizer. Therefore, we conclude that L^* is the unique maximizer of Φ over $[0, 1]$. $\square$

Proof of Theorem 3. Recall from §5.1 that when the firm has rank-based information, its immediate acceptance set A has the property that $\lambda(A) \in \{0, 1\}$. In either case, by Theorem 1, for any $\pi \in \Pi_1^R$, all applicants almost surely choose the same arrival time in equilibrium, independent of the firm's policy and choice of A . This implies that, almost surely, they are ordered uniformly at random. The firm's problem and its optimal policy are identical to the classical secretary problem with $N+1$ applicants; the theorem statement follows from Lindley (1961). $\square$

Proof of Proposition 4. Consider any given firm policy $\pi \in \Pi_1^S$ with immediate acceptance set A . It is necessary that the applicants' best response follows (18). We first consider the case where $\lambda(A) = 1$, in which the firm accepts the first applicant with probability 1. In this case (18) implies that all applicants arrive at time 0 almost surely. The problem therefore reduces to the classical setting with a randomly ordered arrival sequence, where the firm gains strictly more utility by deviating to the decreasing threshold policy characterized in Gilbert and Mosteller (1966).

Next, consider the case that $\lambda(A) < 1$, so that $\lambda(A^c) > 0$. Define a new policy $\pi' = (\pi'_1, \pi'_2, \dots, \pi'_N, 1)$ by setting $\pi'_j = \pi_j$ for all $j = 2, \dots, N$. To define π'_1 , let $F : [0, 1] \rightarrow [0, 1]$, $F(q) := \frac{\lambda(A^c \cap [0, q])}{\lambda(A^c)}$ denote the conditional cdf of a standard uniform distribution restricted to A^c , and construct a measurable set $B \subseteq A^c$ such that

$$[F(q)]^N > \frac{1}{2} \quad \text{for all } q \in B,$$

noting that such B always exists because $\lambda(A^c) > 0$ by assumption. We now define π'_1 as

$$\pi'_1(q) = \begin{cases} 1, & q \in A \cup B, \\ 0, & q \notin A \cup B. \end{cases}$$

We compare the firm's payoff $\mathbb{P}(R_{j_\pi} = 1)$ under π and π' , holding fixed the applicants' arrival behavior in (18). Recalling the decomposition of $\mathbb{P}(R_{j_\pi} = 1)$ from (12) and that for each $k \in \mathcal{N}$, E_k denotes the event that exactly k applicants have scores in A , we have

$$\mathbb{P}(R_{j_{\pi'}} = 1 | E_k) = \mathbb{P}(R_{j_\pi} = 1 | E_k) \quad k = 1, \dots, N+1.$$

This is because on such E_k , k applicants arrive at time 0, and they are each accepted with probability $1/k$ for both policies π and π' . The decomposition (12) implies that

$$\mathbb{P}(R_{j_{\pi'}} = 1) - \mathbb{P}(R_{j_\pi} = 1) = \left[\mathbb{P}(R_{j_{\pi'}} = 1 | E_0) - \mathbb{P}(R_{j_\pi} = 1 | E_0) \right] \mathbb{P}(E_0).$$

Furthermore, because π' and π coincide from period 2 onward, and both policies reject all first applicants whose score is not in B , we have

$$\mathbb{P}\left(R_{j_{\pi'}}^{\dagger} = 1 | E_0\right) - \mathbb{P}\left(R_{j_{\pi}}^{\dagger} = 1 | E_0\right) = \mathbb{P}\left(\left\{R_{j_{\pi'}}^{\dagger} = 1\right\} \cap \{Q_{\sigma(1)} \in B\} \middle| E_0\right) - \mathbb{P}\left(\left\{R_{j_{\pi}}^{\dagger} = 1\right\} \cap \{Q_{\sigma(1)} \in B\} \middle| E_0\right)$$

It therefore suffices to show that the RHS of the above is strictly positive.

On event E_0 , all $N + 1$ applicants have scores in A^c , and they all arrive at time 1. Since B is non-empty, fix an arbitrary $q \in B$, and further condition on the event $\{Q_{\sigma(1)} = q\}$. Under π' , the firm accepts immediately, and

$$\mathbb{P}\left(R_{j_{\pi'}}^{\dagger} = 1 \mid E_0, Q_{\sigma(1)} = q\right) = [F(q)]^N,$$

since success occurs if and only if all remaining N scores are at most q . In turn, this implies that

$$\mathbb{P}\left(\left\{R_{j_{\pi'}}^{\dagger} = 1\right\} \cap \{Q_{\sigma(1)} \in B\} \mid E_0\right) = \int_B [F(q)]^N dF(q),$$

because since quality scores are all randomly ordered, the distribution of $Q_{\sigma(1)}$ conditional on E_0 coincides exactly with F . Conversely, under π , the firm rejects the first applicant because $q \in B \subseteq A^c$. We therefore have

$$\mathbb{P}\left(R_{j_{\pi}}^{\dagger} = 1 \mid E_0, Q_{\sigma(1)} = q\right) \leq 1 - [F(q)]^N,$$

because for the firm to hire the best applicant, it is necessary that at least one of the other N applicants has a score that exceeds q . This implies that

$$\mathbb{P}\left(\left\{R_{j_{\pi}}^{\dagger} = 1\right\} \cap \{Q_{\sigma(1)} \in B\} \mid E_0\right) \leq \int_B 1 - [F(q)]^N dF(q).$$

Therefore,

$$\mathbb{P}\left(\left\{R_{j_{\pi'}}^{\dagger} = 1\right\} \cap \{Q_{\sigma(1)} \in B\} \middle| E_0\right) - \mathbb{P}\left(\left\{R_{j_{\pi}}^{\dagger} = 1\right\} \cap \{Q_{\sigma(1)} \in B\} \middle| E_0\right) \geq \int_B (2[F(q)]^N - 1) dF(q) > 0,$$

where the final strict inequality is by construction of B . Therefore, we have exhibited that π' is a strictly profitable deviation for the firm, which implies that no symmetric BNE exists. $\square$